

Learning Traps

Hassan Sayed*

Princeton University

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Abstract

When constituents are uncertain about whether a leader's policies are aligned with their interests, policy choices generate a tradeoff between payoffs and information. Leaders can implement their preferred policy, but if the outcome is bad, constituents learn this and threaten office removal. I show that, in a repeated setting, the optimal policy of leaders constrained by this threat of information revelation is nonmonotone in beliefs that the leader's policies are aligned with constituents. At extremal beliefs, leaders experiment with the most extreme policy that keeps them in office. At middle beliefs, they implement spatially intermediate "learning trap" policies that halt learning about the optimal policy when it would be most useful. I illustrate applications of the model to the post-1861 reforms ending serfdom in Imperial Russia; as well as political contagion, international conflict, and economic reform.

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1 Introduction

An incumbent's policy choices are shaped not only by her constituents' preferences but also their capacity to remove her from office. Consider a politician who seeks to fund a public good through taxation. If voters value the good, the leader faces minimal resistance and can tax them to efficiently fund the project. If they do not, her choices are constrained: she can only tax voters until they are indifferent to voting her out of office. In both cases, because the effects of policies on constituents are known, the leader implements her most preferred policy that allows her to stay in power.

However, when constituents are *uncertain* about the effects of policies, leaders' policy choices generate not only payoffs but also *information*. Suppose voters do not know whether they will benefit from our politician's public good. The politician could fund the good, and if voters learn they benefit from it, she could maintain her preferred policy. But if they learn otherwise, voters will preclude her ability to fund the good for the rest of her career. Alternatively, she could *partially* fund the good — making it hard for voters to discern whether the outcomes they observe are because the good is valuable and has been funded too little, or because it is not valuable and has been funded too much.

Thus, when constituents are uncertain about what policies benefit them, are leaders incentivized to experiment with their preferred policies? Would a long-lived leader — like a dictator or term-unlimited politician, who may be particularly worried about how her actions today affect what she can accomplish in the future — ever pursue the policies she ideologically champions, or instead seek to limit learning?

This paper answers these questions by developing a theoretical model where an infinitely-lived leader is held accountable by her constituents. The leader has a preferred policy, but her constituents are unsure whether that policy aligns with their interests, and make inferences by observing the effects of the leader's policy choices. The model shows that leaders experiment with their preferred policies only when constituents are relatively certain that their interests are aligned (or misaligned) with the leader's. When constituents are uncertain about what policies benefit them, leaders implement *learning traps*: inefficient, spatially intermediate policies that prevent constituents from inferring the effects of policies. This pattern is a stark contrast to the “gambling for

resurrection” logic ubiquitous in the political agency literature — where short-lived leaders “gamble” with their preferred policies when they are likely to be misaligned with their constituents and otherwise pursue learning traps.¹

Theoretical Overview In the model, a leader prefers a higher policy $x \in [0, 1]$. A representative for “the people” wants to maximize an outcome y , and can threaten to overthrow the leader. There are two states of the world: a “pro-leader” state where the relationship between x and y is positive — meaning the leader’s preferred policy $x = 1$ is optimal — and an “anti-leader” state where it is negative, meaning $x = 0$ is optimal. The leader and people share a common prior belief. As different policies are implemented, information is revealed. The leader and people use histories of policy-outcome data (y, x) to infer the state. The key feature is that data on policy efficacy are generated endogenously: by setting x , the leader also controls inference. Information revolves around an intermediate “learning trap” policy x_{LT} which, when implemented, generates no information about what the optimal policy is.

The intuition for the main prediction of the model — that leaders implement learning traps when beliefs in the “pro-leader” state are intermediate — is as follows. In a repeated setting, a leader chooses between implementing the learning trap for all time — perpetually shutting down information but receiving a modest payoff — or experimenting with her preferred policy. Experimentation could work in the leader’s favor — in which case she can forever pursue her preferred policy — but could also backfire — in which case the people will force a low policy. When constituents believe the pro-leader state is more likely, revealing information favors the leader in expectation, meaning she has an incentive to experiment with a high policy. When constituents are unsure about the state, the leader implements x_{LT} — since experimentation could tie her hands if the anti-leader state materializes. When constituents believe the anti-leader state is more likely, the leader would like to halt learning, but constituents threaten overthrow and force her to pursue a very low policy.

The assumption of a patient, long-lived leader is crucial for the result, and drives the present paper’s differences leader with the “gambling for resurrection” literature. In her last period in

¹See Downs and Rocke (1994) and Majumdar and Mukand (2004) for applications to war and policy experimentation, and Miller (2005) for an overview of applications to principal-agent models in politics.

office, a short-lived leader would pursue the highest policy that prevents overthrow. If the pro-leader state is more likely, she would prefer to halt learning with x_{LT} to ensure she can implement a high policy tomorrow. If the anti-leader state is unlikely, the leader pursues a policy farther from x_{LT} in the hopes it will reveal information that favors her.

I show that the model’s basic predictions still hold even if the people have a preference for learning about the state of the world. Even if the people are “self-aware” that their learning has been halted, the leader still pursues policies that *minimizes* information. I additionally discuss how the model’s findings are robust to alternative information structures, more than two states of the world, uncertainty about (or heterogeneity in) constituent beliefs, and exogenous forms of information revelation.

Applications to Land Reform in Russia and Political Contagion Can we use the insights of the model to learn about polities’ unwillingness to reform in the face of uncertainty? To illustrate how the mechanisms of the model can influence the policy choices of autocratic regimes, I study how “learning traps” played a role in the 1861 reforms ending serfdom in Russia. Prior to 1861, Russian peasants (serfs) worked noble land as part of communes. After Russia’s 1856 defeat in the Crimean War, many in the Tsar’s government believed liberal reforms — like private property and freeing labor to move to the cities — could stimulate economic growth, reduce the power of the nobility (Moon, 2014) jumpstart an industrial revolution (Pereira, 1980), and increase fiscal revenue (Dennison, 2020, 2023). However, they worried liberal policies could backfire, leading to exploitation, abandonment of land, fiscal straits, and overthrow (Dennison, 2014; Polunov et al., 2015). That is, there was uncertainty about whether the government’s preferred policy was in the interest of constituents.

In the ensuing reforms, the government pursued a middle-ground: it mixed elements of private with communal decision-making — distorting investment and production decisions (Dennison, 2020), restricted labor mobility, and inhibited urban migration and industrial development (Nafziger, 2010). I argue that this arrangement illustrates the logic of a learning trap: the government feared unrest if liberal policies backfired, but by pursuing an intermediate policy that mixed liberal and communal institutions, distorted labor and production decisions to halt learning about which policy was better.

Extending beyond Russia, can the model provide insight into regional variation in policy experimentation? I extend the model with *exogenous external information revelation* so that leaders' policy decisions are not the only sources of information about the effects of policies. I show that leaders pursue learning trap policies for a smaller range of beliefs: they cannot totally shut down learning and hope experimentation with their preferred policy will favor them. I use this extension to study political "contagion," where groups of countries with similar historical institutions learn from each other. I show how these predictions can explain regional variations in policy experimentation: in particular, why Europe experienced more experimentation with indirect rule than China in the 19th century. I finish by discussing applications of learning traps to the USSR, healthcare reform in FDR's New Deal, and the informational externalities of "limited war."

Plan of Paper I introduce the model in section two and solve it in section three, where I also discuss theoretical extensions. Section four uses the end of Serfdom in Russia to illustrate how a patient regime may pursue a learning trap. Section five solves the model with exogenous information revelation to study political fragmentation in Europe and China. Section six addresses additional applications of learning traps, and I conclude in section seven. The remainder of the section discusses how the paper contributes to our understanding of the theoretical literature on political accountability and voter learning.

Theoretical Contributions This paper's central insight is that leaders pursue learning traps at *intermediate* beliefs that their preferred policy favors their constituents. This result contrasts with a popular result in the political accountability literature known as "gambling for resurrection," where leaders pursue their preferred policy at lower beliefs and learning traps at higher beliefs (Prendergast and Stole, 1996; Dur, 2001; Majumdar and Mukand, 2004; Fu and Li, 2014; Dewan and Hortala-Vallve, 2019; Tomasi, 2023; Izzo, 2024). This difference is driven by two assumptions. First, the present paper's leader is *infinitely-lived*. Short-lived leaders always pursue their preferred policies in their last period in power. At high beliefs, halting learning thus guarantees leaders can pursue their preferred policies tomorrow. In repeated settings, leaders choose between a learning trap for all time — generating a perpetually moderate payoff — or a one-off extreme policy that reveals information. Information revelation will only be preferred to a learning trap if it delivers

more than a moderate payoff in expectation, which only happens at high beliefs.

Second, leaders in “gambling for resurrection” models can often be replaced by outside options. The only way for leaders to secure retention at low beliefs is to experiment with their preferred policy and hope for a strong signal. In the present paper, constituents’ overthrow threats bind saliently only when they are *sure* that the leader’s preferred policy is bad, providing flexibility to pursue learning traps. The present paper’s insights are conceptually related to partisan traps (de Mesquita and Dziuda, 2024), where voters perpetually elect partisan ideologues who never pursue common value policies. However, “partisan traps” arise due to information asymmetries that obfuscate the feasibility of non-partisan policies.

Because of its analysis of a repeated environment, the paper contributes to a theoretical literature on the reputational motives of patient agents. Holmström (1999) shows that managers oversupply effort to *slow* learning when firms set future wages based on inferences about their ability.² Non-learning results in these models often depend on asymmetries in costs (Manso, 2011), information (Ely and Välimäki, 2003), or signal interpretation (Aghion and Jackson, 2016). In the latter, constituents threaten to replace incumbents until a high-ability leader emerges, forcing learning in equilibrium. However, this mechanism is driven by a “gambling for resurrection” phenomenon which cannot be replicated in the present paper. This paper finds more patient leaders prefer to slow learning since the salience of *future* constraints on policies increases, the opposite of Besley and Case (1995) and Banks and Sundaram (1998).

This paper adds to our understanding of how voters using endogenously generated policy-outcome data may learn inefficiently (Levy, 2014; Spiegler, 2016; Eliaz and Spiegler, 2020; Levy and Razin, 2021; Schwartzstein and Sunderam, 2021; Montiel Olea et al., 2022). Voters learning from spatial platforms may converge on inefficient policies (Callander, 2011). Frequentist inference or bounded memory can cause polarization (Izzo et al., 2021; Levy et al., 2022; Levy and Razin, 2025). Less information about policies can be better for voters (Prat, 2005) or politicians (Kartik et al., 2015). Substitutability between politician effort and ability (Ashworth et al., 2017) can hamper learning about ability. Empirical work also documents inferences about policy efficacy, including trust in government (Chen and Yang, 2019), external validity of policy experiments (Wang and Yang, 2021), and policy contagion (Buera et al., 2011; Mukand and Rodrik, 2005).

²Crucially, learning still occurs in Holmström’s model; non-learning requires an evolving state.

A common theme throughout models of autocratic power-balancing is that rulers self-regulate their exploitation in order to avoid overthrow; they implement the most extreme policies keeping constituents indifferent to overthrowing them (Acemoglu and Robinson, 2000, 2001; De Mesquita and Smith, 2010; Boix and Svolik, 2013; Li et al., 2014; Dower et al., 2018).³ This paper argues that leaders *moderate* their actions when policies generate information. It also relates to gradualism in “divide-the-dollar” settings like land reform, as discussed in Roland (2002) and Acemoglu and Robinson (2008), showing that under uncertainty about the optimal form of land ownership, “partial reforms” mixing different property regimes can obfuscate which reform is better.

2 Model

Policies Consider an infinite-horizon model beginning at $t = 1$. Each period, a policy $x_t \in X = [0, 1]$ is implemented which determines an outcome $y_t \in \mathbb{R}$ realized later that period. For example, y can represent profits, income, or the value of a public good.

Agents The first agent in the model is the “leader” — an entrenched incumbent, autocrat, or CEO. The second agent — “the people” — constrains the leader’s power and can be interpreted as a representative for an electorate, board of directors, or elite group. Both agents are infinitely lived.

The leader discounts the future at rate δ . Her flow utility is $u_\ell(x_t)$ for $u_\ell(\cdot)$ weakly concave and strictly increasing. Preference for higher x_t could reflect an ideological bent; reputational concerns; or the size of rents she can extract in a setting like land reform.

The people’s flow utility is y_t . They discount the future at rate 0. This simplifies the model to illuminate its key tradeoffs, but we relax this assumption in section 3.2.

Actions At $t = 1$, the leader holds power. At the beginning of each period she is in power, the leader announces a policy. The people decide whether to incur a one-time cost $c > 0$ to overthrow her. If they overthrow, the people set x_t from t onwards. Overthrow succeeds with probability 1; the leader receives a large loss $-L < 0$ and no further flow utility. If the leader is not overthrown,

³Dower et al. (2018) looks at authority given to peasant communes given the frequency of disruptions in 1860s Russia, the period of this paper’s case study.

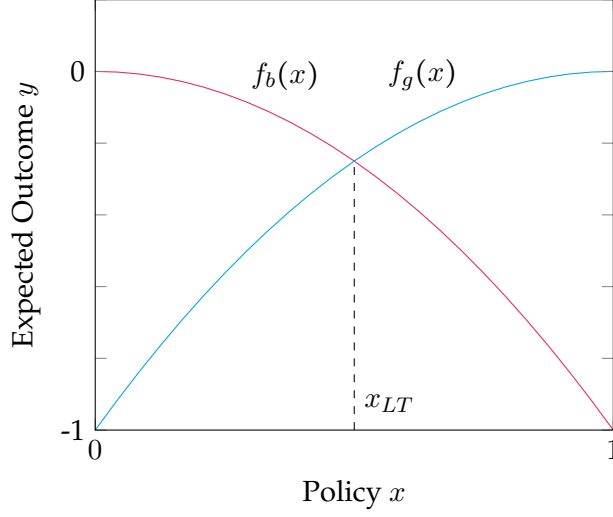


Figure 1: $f_g(x)$ and $f_b(x)$ graphed

her announced policy x_t is implemented that period and t moves to $t + 1$.

States Agents possess uncertainty over two states of the world governing the relationship between x and y . In the “pro-leader” state, increases in x cause increases in y . In the “anti-leader” state, increases in x cause decreases in y :

$$y_t = -(x_t - 1)^2 + \epsilon_t \equiv f_g(x_t) + \epsilon_t \quad \text{Pro-Leader} \quad (1)$$

$$y_t = -x_t^2 + \epsilon_t \equiv f_b(x_t) + \epsilon_t \quad \text{Anti-Leader} \quad (2)$$

$\epsilon_t \sim \mathcal{U}[-\sigma, \sigma]$ is i.i.d. every period, making inference potentially imperfect.⁴ f_g and f_b are graphed in Figure 1.

The expected value of y conditional on $x = 1$ is higher in the pro-leader than the anti-leader state, and $x = 0$ is better in the anti-leader than the pro-leader state. This payoff symmetry means that $f_g(x) = f_b(x)$ at a policy $x_{LT} = \frac{1}{2}$ with $y_{LT} \equiv f_g(x_{LT})$.

Inference Both agents follow Bayes’ Rule and use histories $\{y_\tau, x_\tau\}_{\tau \leq t}$ to update a common belief over the two states. Denote q_{t-1} the belief in the pro-leader state at the beginning of time t . During period t , agents observe a policy-outcome pair (y_t, x_t) and use it to update q_{t-1} to q_t , to be carried

⁴Proposition 3 shows the results are robust to single-peaked distributions of the noise terms.

into $t + 1$. Denote $\Delta(x) = |f_g(x) - f_b(x)|$ the *difference* between the expectation of y conditional on x in each state.

For ease of mathematical exposition, we assume $\sigma \geq \frac{1}{2}$. Because $|f_g(x) - f_b(x)| \leq 1$, $\frac{\Delta(x)}{2\sigma}$ is bounded above by 1. Bayes' Rule implies that the distribution of posteriors conditional on x_t and q_{t-1} is:

$$q_t | q_{t-1}, x_t = \begin{cases} 1 & \text{with prob. } \frac{\Delta(x_t)}{2\sigma} q_{t-1} \\ q_{t-1} & \text{with prob. } 1 - \frac{\Delta(x_t)}{2\sigma} \\ 0 & \text{with prob. } \frac{\Delta(x_t)}{2\sigma} (1 - q_{t-1}). \end{cases}$$

When $x_t = x_{LT}$, $f_g = f_b$, meaning $\Delta(x_{LT}) = 0$. x_{LT} shuts down information revelation ($q_{t-1} = q_t$ with probability 1) and we refer to it as a **learning trap policy**.⁵ While extreme policies reveal more information, x_{LT} is a “muddled policy” that mixes elements of these opposing extremes to hamper inference.

We make two more assumptions. First, we normalize $u_\ell(1) = 1$ and $u_\ell(x_{LT}) = x_{LT}$. Second, we assume that $c < x_{LT}^2$. This upper bound on the cost of overthrow ensures that if the anti-leader state realizes, the leader cannot implement moderate policies like x_{LT} without risking overthrow.

Sequential Game We describe the game recursively.

1. At time t , if in power, the leader announces a policy x_t . Then, the people decide whether to overthrow the leader.

(a) If the people decide to overthrow the leader:

- i. The people incur the cost of overthrow c . The leader loses power, receives $-L$, and receives no further flow utility.

- ii. The people set x_t , y_t realizes, and they receive utility y_t .

⁵The existence of x_{LT} only depends on the single-crossing property of f_g and f_b , and generalizes beyond uniform noise. If $\epsilon_t \sim p(\epsilon_t)$ for some distribution p :

$$\begin{aligned} q_t &= \frac{p(y_t - f_g(x_{LT}))q_{t-1}}{p(y_t - f_g(x_{LT}))q_{t-1} + p(y_t - f_b(x_{LT}))(1 - q_{t-1})} = \frac{p(y_t - f_g(x_{LT}))q_{t-1}}{p(y_t - f_g(x_{LT}))q_{t-1} + p(y_t - f_g(x_{LT}))(1 - q_{t-1})} \\ &= \frac{p(y_t - f_g(x_{LT}))q_{t-1}}{p(y_t - f_g(x_{LT}))} = q_{t-1} \end{aligned}$$

- iii. The people update q_{t-1} to q_t . Return to step 1(a)ii at $t + 1$.
- (b) If the people decide not to overthrow the leader:
 - i. The leader's policy x_t is implemented, y_t realizes, and the people receive utility y_t .
The leader in power receives $u_\ell(x_t)$.
 - ii. Agents update their belief q_{t-1} to q_t . Returns to step 1 at $t + 1$.

We focus on pure-strategy Markov Perfect Equilibria with respect to the belief q_{t-1} . $x_\ell(q) \in [0, 1]$ indicates the leader's policy plan. The people's strategy is given by $\{p(q), x_p(q)\}$. $p(q) \in \{\text{overthrow}, \text{nooverthrow}\}$ denotes the overthrow decision. $x_p(q) \in [0, 1]$ indicates their policy plan once in power. $x_\ell(q)$ and $\{p(q), x_p(q)\}$ must best respond to each other in equilibrium.

Comments The model lends itself to two interpretations of information. The first is policy uncertainty: neither the leader nor people know the effects of policies until they are implemented. The second is that the people utilize one of two subjective models of the world to evaluate outcomes: a pro-leader data-generating process ($y_t = f_g(x_t) + \epsilon_t$) and anti-leader process ($y_t = f_b(x_t) + \epsilon_t$). Section 3.3 shows that the model's predictions are similar if constituents' beliefs are heterogeneous (or unknown) using a framework of *ambiguity*.

That learning trap policies are *spatially intermediate* is driven by the assumption that the people's maximal payoff in each state of the world is identical, qualitatively suggesting that learning traps mix elements of extreme policies to slow learning. What will matter for the results is that the leader's equilibrium payoff in the pro-leader state is higher than in the anti-leader state, and that her utility from the *learning trap* is in between.⁶ Thus, even if learning trap policies were *extreme* (e.g. $x = 0$), endowing the leader with a reputational loss for pursuing a failed policy could deliver similar results.

Another plausible variant of the model could be a "linear regression format," where payoffs are of the form $y_t = \beta_0 + \beta_1 x_t + \epsilon_t$, and the people believe (β_0, β_1) could take any value in $\mathbb{R}^2$. Here, the logic of a learning trap could be replicated through *policy variation*. As in Levy and Razin (2025), the people quickly learn the state of the world if there is a great deal of policy variation.

⁶The insights of the model — and proofs of the main results — are identical if f_g and f_b are *linear* in x with positive/negative slope.

Learning is limited if there is no variation, meaning a leader who repeatedly implements the same policy could minimize information revelation.

3 Analysis

We analyze the baseline model in Section 3.1, highlighting the forces predicting why the leader may pursue a learning trap only at intermediate beliefs in the pro-leader state. Section 3.2 solves a general version of the model where the people value both current and future payoffs — showing that even if the people are “self-aware” that their learning has been shut down, the leader would still want to minimize information. Section 3.3 discusses further extensions to the model: information asymmetry, alternative information structures, and ambiguity about constituents’ beliefs.

3.1 Analysis of Baseline Model

People The people’s optimal policy, conditional on being in power, is $x_p(q) = \arg \max_x \mathbb{E}[y|x]$; expectations are taken with respect to q . The people have a strict incentive to overthrow if and only if

$$\max_x \mathbb{E}[y|x] - c > \mathbb{E}[y|x_\ell(q)] \quad (3)$$

i.e. if the best policy they can achieve is better than the leader’s, less the cost of overthrow. In equilibrium, they overthrow if and only if the inequality above is satisfied strictly. The people’s optimal policy maximizes $-qx^2 - (1-q)(1-x)^2$, which occurs when $x_p(q) = q$.

For each q , we define $\text{NR}(q)$ as the set of policies that weakly prevents overthrow: $\text{NR}(q) = \{\hat{x} \in [0, 1] : \max_x \mathbb{E}[y|x] - c \leq \mathbb{E}[y|\hat{x}]\}$. $\text{NR}(q)$ is monotone in q , in the sense that its minimal and maximal elements are increasing in q . Because x_{LT} delivers the same utility in both states of the world, there exist a range of interior beliefs where x_{LT} is contained in the interior of $\text{NR}(q)$. Its graph as a function of q is shaded below. The optimal policy for the people $x_p(q)$ is bolded.

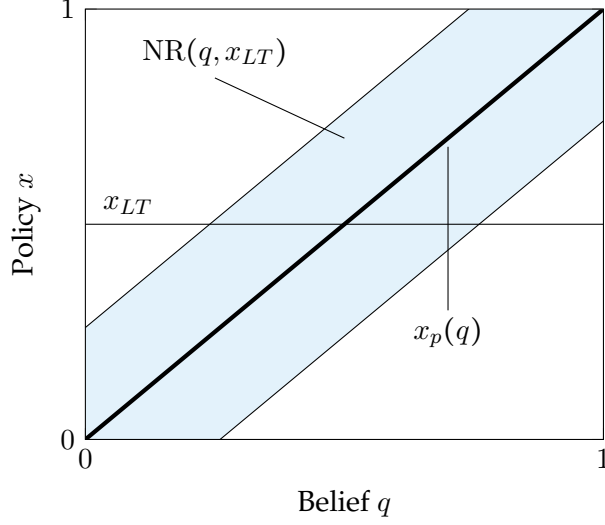


Figure 2: No-Overthrow Constraint $NR(q)$

Leader's Solution Denote by $\bar{x}(q) \equiv \max\{NR(q)\}$ the most extreme policy avoiding overthrow. At $q = 0$ or 1 , policies have no information externalities, so the leader implements $\bar{x}(q)$. In particular, if $q = 1$, the leader can implement $\bar{x}(1) = 1$ for all time, and her payoff is $u_\ell(1) = 1$. At $q = 0$, the leader implements $\bar{x}(0) > 0$ for all time. Because $c < x_{LT}^2$, $\bar{x}(0) < x_{LT}$. We denote $u_\ell(\bar{x}(0)) = \underline{u}$. The key assumption for the main result is that $1 > x_{LT} > \underline{u}$: the leader's utility is higher than x_{LT} at $q = 1$ and lower at $q = 0$.

Since L is large, the leader will avoid overthrow on the equilibrium path. For interior q , the leader's optimal policy $x_\ell(q)$ is the solution to the following Bellman Equation:

$$V(q) = \max_{x \in NR(q)} \underbrace{u_\ell(x)}_{\text{Flow utility}} + \delta \left(\underbrace{\frac{\Delta(x)}{2\sigma} \Psi(q)}_{\text{Truth revealed}} + \underbrace{\left(1 - \frac{\delta \Delta(x)}{2\sigma}\right) V(q)}_{\text{Truth not revealed}} \right) \quad (4)$$

where $\Psi(q) = q + (1 - q)\underline{u}$ is the *expected value from revealing the truth*. Implementing a policy x generates information in addition to flow utility. With probability $q\Delta(x)/2\sigma$, $q \rightarrow 1$. With probability $(1 - q)\Delta(x)/2\sigma$, $q \rightarrow 0$. In the former case, the leader's value function is 1 and in the latter it is $\underline{u}$. Otherwise, q is invariant.

The following proposition characterizes $x_\ell(q)$ for patient leaders.

Proposition 1. *Suppose $\sigma(1 - \delta)$ is small or $\underline{u}$ is sufficiently low. There exists a threshold $\underline{q} \in (0, 1)$ such that:*

1. If $q \leq \underline{q}$, $x_\ell(q) = \min\{x_{LT}, \bar{x}(q)\}$. The leader plays the learning trap or the policy closest to it.
2. If $q > \underline{q}$, $x_\ell(q) = \bar{x}(q)$. The leader plays the maximal policy preventing overthrow.

Proof. For $q \in (0, 1)$, suppose the leader is constrained at $q = 0$ or 1 but can otherwise implement any policy in $[\underline{x}(0), 1]$. We can rearrange the leader's Bellman equation as:

$$\max_{x \in [\underline{x}(0), 1]} \frac{2\sigma(1-\delta)}{2\sigma(1-\delta) + \delta\Delta(x)} u_\ell(x) + \frac{\delta\Delta(x)}{2\sigma(1-\delta) + \delta\Delta(x)} \Psi(q) \quad (5)$$

This expression is a convex combination of flow utility $u_\ell(x)$ and the value $\Psi(q)$ of the truth. If the leader pursues a strategy where she implements x_{LT} for all time, it delivers a value $u_\ell(x_{LT}) = x_{LT}$. As x moves away from x_{LT} , weight shifts from flow utility $u_\ell(x)$ onto the expected value of revealing information $\Psi(q)$.

Suppose q is low, so that $\Psi(q) < x_{LT}$. If the leader implements $x > x_{LT}$, this results in higher flow utility relative to x_{LT} . But it also reveals information, whose expected value $\Psi(q)$ is worse than x_{LT} . If $\sigma(1-\delta)$ is small, the effect of flow utility is small, meaning that pursuing the learning trap for all time and receiving a modest payoff is better than gaining a bit of flow utility that reveals unfavorable information, up to a point $\underline{q}$.

If $\Psi(q) \geq x_{LT}$, information is better in expectation than shutting down learning and receiving a moderate payoff. Pursuing $x > x_{LT}$ both increases flow utility and the weight on the revelation of favorable information. For high q , because information is favorable, $x < x_{LT}$ may counterintuitively be preferred to x_{LT} after a point $\bar{q}$.

The solution to this *unconstrained* problem is shown in the left panel of Figure 3 in red. Crucially, the vertical axis in this figure represents the *utility space* of the leader. We let $\underline{u} < 0$ so that $\Psi(q)$ is below 0 for low q . The arrows reflect the direction of the leader's *preference* in the unconstrained problem, which tends to the learning trap below $\underline{q}$ and extreme policies above $\bar{q}$.⁷ Below $\underline{q}$, the leader prefers the learning trap for both flow utility and informational purposes; between $\underline{q}$ and $\bar{q}$, prefers a high policy for flow utility purposes; and above $\bar{q}$, prefers any extreme policy to x_{LT} for informational purposes.

The right panel transposes the constraint set $\text{NR}(q)$ (shaded) onto the *policy space*.⁸ By overlay-

⁷Formally, the arrows show the sign of the derivative in this unconstrained problem

⁸A more rigorous proof of this argument, as well as Proposition 1, is presented in the Online Appendices Lemma 8

ing the direction of the leader's preference from the unconstrained problem onto this constraint set, we trace out the leader's solution $x_\ell(q)$ in the constrained problem (bolded in red), which forces the leader to pursue $x < x_{LT}$ for low q .

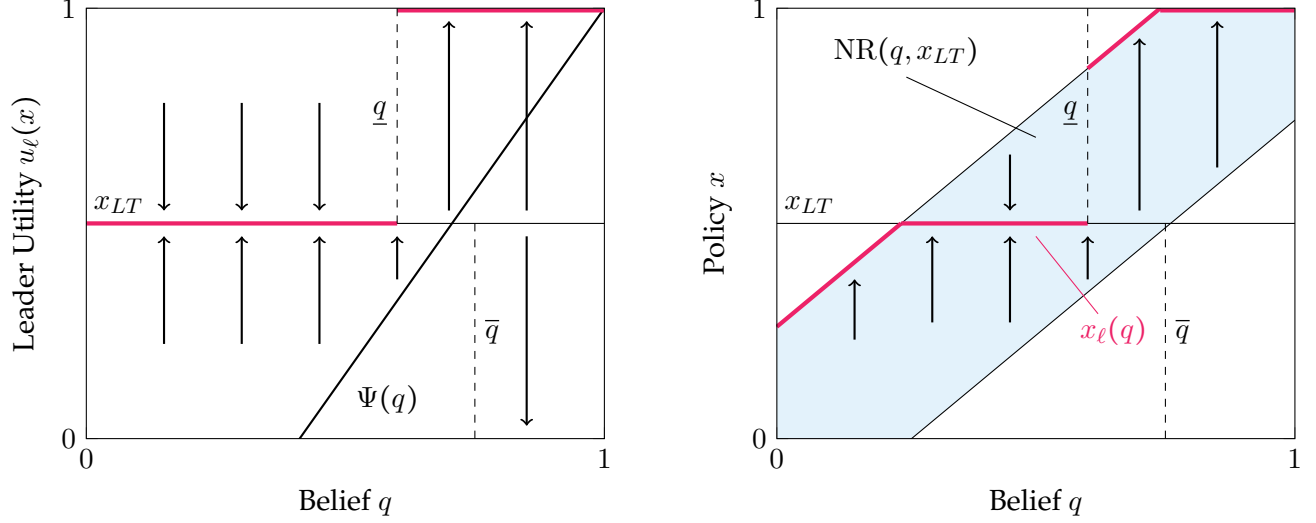


Figure 3: Leader's Policy Preferences in Unconstrained Leader's Problem (Left) vs. Leader's Constrained Solution (Right)

□

Proposition 1 highlights a nonmonotonicity in policy experimentation. When the optimal policy is relatively certain, leaders pursue extreme policies. For high q , they implement their most preferred policies, and for low q , the highest policy keeping the people indifferent to overthrow. If the optimal policy is uncertain, leaders moderate policies to suppress information that could constrain their actions in the future — precisely when learning about the optimal policy would be most valuable. Notably, the people have a strict incentive to *retain* the leader at intermediate q , a contrast to dictator games where leaders always keep the people indifferent to overthrow. The result also yields the following comparative statics.

Proposition 2. *Suppose σ decreases, δ increases, or $\underline{u}$ decreases. Then $\underline{q}$ increases.*

As σ decreases, for each x , the probability of information revelation increases. As δ increases, the leader deemphasizes flow utility; $(1 - \delta)$ could represent the length of the leader's office terms or the frequency of evaluations. In both cases, the leader is worried more about how her actions and Theorem 1.

may reveal the truth, increasing attraction to learning traps. As $\underline{u}$ decreases, for all q , the value of revealing information decreases.

Policy Dynamics Figure 3 provides insights into policy variation due to inheritance of status quo policies or exogenous shifts in beliefs. Suppose a leader inherits a status quo “out-of-equilibrium” policy $x_0 = 1$ and belief q_0 . If q_0 is intermediate, the leader will swing from $x_0 = 1$ to x_{LT} . If q_0 is low, the leader will shift to $\bar{x}(q_0)$, which is the closest policy to x_{LT} that averts overthrow. If q_0 is high, she will sustain the status quo $x = 1$. Thus, despite the fact that $x = 1$ is ideologically favorable to the leader, at intermediate beliefs, she may shift to a less-desirable policy because of its informational features.⁹

Conversely, suppose the leader is already pursuing her optimal policy $x_\ell(q_0)$, but that q_0 shifts to q'_0 (and this shift is *not anticipated* by the leader).¹⁰ Suppose q_0 is intermediate so that $x_\ell(q_0) = x_{LT}$. If q'_0 is slightly above or below q_0 , $x_\ell(q'_0) = x_{LT}$ as well; the leader’s policy is *not sensitive* to small shifts in beliefs. If q'_0 is close to 0, policy shifts towards $x = 0$; the leader would like to pursue x_{LT} but is forced into a lower policy by the no-overthrow constraint. If q'_0 is closer to 1, however, the leader’s policy swings towards $x = 1$. Thus, beginning at intermediate beliefs, small shocks to beliefs generate no policy variation. Larger shocks favoring the pro-leader state can generate swings to the leader’s preferred policy, while those favoring the anti-leader state generate less dramatic shifts to lower policies.

Comparison to Short-Lived Leader and Term Limits How does the behavior of this infinitely-lived leader compare to a leader who only lives two periods? In a two-period model, a leader would always implement the highest possible policy in the second period, $\bar{x}(q)$. Hence, the value of pursuing x_{LT} would not be x_{LT} for all time, but x_{LT} today and $\bar{x}(q)$ tomorrow. At high beliefs, this would motivate her to halt learning.

This comparison illuminates policy differences between “short-lived” and “long-lived” governments. Short-lived, term-limited politicians only implement their preferred policies when they need to “gamble” to save their chances of retention, but may pursue their preferred policy right before leaving office. Long-lived politicians — such as persistent dictatorships — pursue their

⁹The next section solves the model when the people value future payoffs *and* may be averse to policy adjustments.

¹⁰Section 5 solves the model when shifts are *anticipated*.

preferred policies if they are likely to be aligned with constituents' interests, but may otherwise only be willing to pursue learning traps.

Comparison to Model with Outside Option Many political agency models study how a leader makes decisions when her constituents are uncertain about her competence. If constituents in these models believe their leader is incompetent, they can replace her with an outside challenger. The basic learning trap results would not hold if the people received a constant outside option upon overthrow. Suppose x_{LT} were preferred to the outside option at $q = 1/2$. Because the people's utility from x_{LT} is same in both states of the world, at $q = 0$, x_{LT} would still be preferred to the outside option. The leader could then achieve utility at least x_{LT} at $q = 0$, eroding the need to implement x_{LT} at higher beliefs. The model then contrasts situations where constituents are uncertain about idiosyncracies of their leader's governance with ones where constituents are uncertain about a *policy fundamental*. Their ability to discipline the leader's policy choices is a function of their information, and is weakest when they have the least amount of information.

3.2 Model with Preference for Learning

We have so far assumed the people value only current-period payoffs. However, if they value future payoffs — and thus learning — can they erode patient leaders' willingness to minimize information? This section establishes that the fundamental insights of the model extend to this more general setting: the leader will *minimize* learning at intermediate beliefs, and will be able to implement the learning trap itself if the *people* suffer policy adjustment costs.

We now assume the people discount the future at rate $\delta > 0$. The leader discounts the future at rate δ^ℓ . The people bear adjustment costs $\kappa|x_t - x_{t-1}|$, capturing frictions from institutional reform or costly experimentation.¹¹ Their utility is $u_p(y_t, x_t, x_{t-1}) = y_t - \kappa|x_t - x_{t-1}|$. We assume that $|f'_g(x_{LT})| = |f'_b(x_{LT})| > (1 - \delta)\kappa$, which ensures that policy adjustment is not *so* costly that policy never moves. We calculate Markov Perfect Equilibrium with respect to the prior q and yesterday's policy x' . We write $x_p(q, x')$ as the people's optimal policy conditional on overthrow.

¹¹Forcing the leader to bear these costs further strengthens the result.

Solution to People’s Problem The solution to this more general version of the model is technically involved, and we preview the results here while delegating the formal proofs to the “Proofs of Results with Patient People.” First, we solve the people’s new problem. Their optimal policy $x_p(q, x')$ solves the following equation:

$$\begin{aligned}
W(q, x') = \max_{x \in [0,1]} & \underbrace{\mathbb{E}[y|x] - \kappa|x - x'|}_{\text{Flow utility}} \\
& + \delta \left(\underbrace{\frac{\Delta(x)}{2\sigma} (qW(1, x) + (1 - q)W(0, x))}_{\text{Truth revealed}} + \underbrace{\left(1 - \frac{\Delta(x)}{2\sigma}\right)W(q, x)}_{\text{Truth not revealed}} \right)
\end{aligned} \tag{6}$$

The first term represents flow utility, and the second set of terms represents their continuation value, which depends on the information that policies reveal.

Proposition 3 in the “Proofs of Results with Patient People” characterizes the solution to the equation, which is displayed in Figure 4 and can be described as follows. Suppose the status quo policy is x_{LT} . For intermediate q , the people face a tradeoff: extreme x generates information — which is now incredibly valuable. But because there is uncertainty about whether high or low x is optimal, they may incur excessive adjustment costs by experimenting in one direction, learning optimal policy is in the other direction, and then “backtracking.” The solution is discontinuous around $q = \frac{1}{2}$, where experimenting in either direction is risky; they pursue policies closer to the status quo x_{LT} , although never x_{LT} itself, since this would preclude learning. There is no tradeoff at extremal q : there is relative certainty about the effects of policies, meaning backtracking costs are not likely to be incurred. Thus, the people pursue extreme policies. This solution is **bolded** below, where the status quo policy x' is calibrated to x_{LT} .

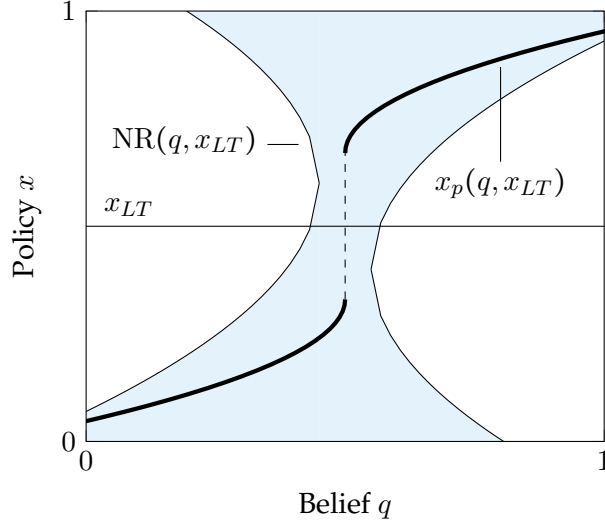


Figure 4: Optimal Policy $x_\ell(q, x_{LT})$ and No-Overthrow Constraint $NR(q, x_{LT})$ when People Value Learning

The figure additionally graphs $NR(q, x')$, the set of feasible *strategies* the leader can implement at belief q and status quo x' . Notably, the set $NR(q, x')$ may not be a closed interval, fixing q and x' . Intuitively, a preference for learning translates to a preference for extreme policies. This means the people may permit the leader to implement extreme policies that deliver suboptimal flow utility. If adjustment costs are high, there exists a set of (q, x') values where the leader can implement a learning trap x_{LT} and avert overthrow.

Leader's Problem The leader's problem is similar to that in equation (4), with the subtlety that she must condition her choices on the status quo x' . At $q = 0$ or 1 , the leader will pursue the most extreme policy preventing overthrow. The equation describing her optimal solution $x_\ell(q, x')$ is then:

$$V(q, x') = \max_{x \in NR(q, x')} u_\ell(x) + \frac{\delta^\ell \Delta(x)}{2\sigma} \Psi(q, x) + \delta \left(1 - \frac{\delta^\ell \Delta(x)}{2\sigma}\right) V(q, x) \quad (7)$$

where $\Psi(q, x')$ represents the *expected value* of revealing information. As before, we let $\underline{u}(x')$ describe the leader's expected utility conditional on the anti-leader state realizing given status quo x' . The following Theorem describes $x_\ell(q, x')$.

Theorem 1. Suppose $\sigma(1 - \delta^\ell)$ is small or $\underline{u}(x')$ is sufficiently low. Then, there exist thresholds $\underline{q} < \bar{q} \in$

$(0, 1)$ such that the following hold:

1. If $q \leq \underline{q}$, the leader plays the closest policy to the learning trap (which may be greater or less than x_{LT}).
2. If $q \in [\underline{q}, \bar{q}]$, the leader plays either the lowest policy below the learning trap or her highest feasible policy.
3. If $q > \bar{q}$, the leader either plays her highest feasible policy or lowest feasible policy.

The central forces of the model here are identical to the baseline: leaders prefer learning traps at intermediate and low beliefs and their preferred policy otherwise. The key difference in equilibrium behavior is driven by the shape of the no-overthrow constraint. At low beliefs, the people may not allow the leader to implement x_{LT} , but may allow her to implement high policies because of their informational value. Conditional on being forced to reveal some information, leaders may prefer these high policies. These potential nonmonotonicities are reflected in Figure 5, which replicates the intuitions of Figure 3. As before, the arrows display the sign of the leader's *preference*, derived from the leader's unconstrained problem.

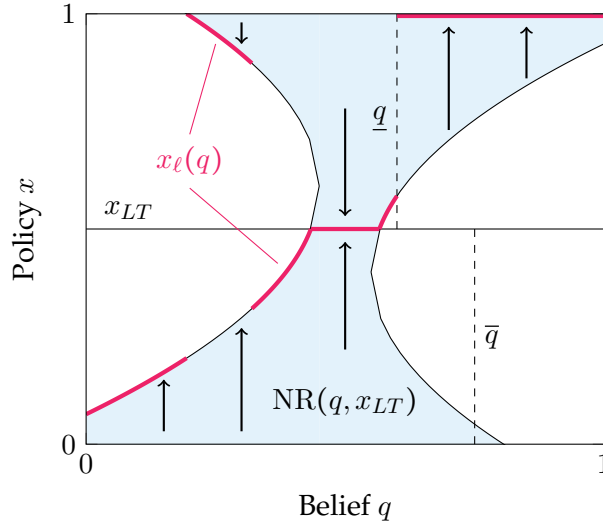


Figure 5: Leader's Constrained Solution when People Value Learning

3.3 Additional Extensions

We now discuss other extensions to the benchmark in section 3.1: information asymmetry, alternative noise specifications, multiple states of the world, and heterogeneous (or uncertain) constituent

beliefs. Proofs of formal results are in the Online “Additional Results.”

Information Asymmetry The baseline model assumes the leader and people possess symmetric information about the state of the world. However, leaders may have more access to information about policies and their effects (e.g., through the work of the public service). How would asymmetric information change the model’s insights?

Suppose the leader receives a private signal about the state. The leader uses her *private* information to assess the value of revealing information, while her actions are constrained by *public* information via the no-overthrow constraint. Suppose we begin at an intermediate belief where the leader pursues x_{LT} . If the signal weakly favors the pro-leader state, the leader may still want to pursue x_{LT} . If the signal favors the anti-leader state, she will surely want to pursue x_{LT} . The leader’s policy hence does not change *and* the people cannot infer her signal from her actions, meaning the model’s predictions are as in the baseline. However, if a pro-leader signal is strong enough to change the leader’s policy choice (e.g. from x_{LT} to $x = 1$), the leader’s actions also reveal her signal: $x = 1$ means the leader received a positive signal, while x_{LT} means she received a negative signal, which shifts her down the no-overthrow constraint.

Possessing a private signal thus creates additional risk for the leader, and she may be better off *without* the burden of private information. If, at the beginning of time, the leader could choose whether to set up an *institution* to acquire private information, she may choose not to. Consequently, long-lived leaders may be averse to setting up systems that assess the effects of experimental policies beyond what constituents observe.

Model with Non-Uniform Noise Are the insights of the model robust to alternative distributions of the noise shock ϵ_t — such as a normal distribution? Under a technical condition, the result generalizes. Suppose $\epsilon_t \sim p$, where $p(\cdot)$ is continuous, has mean 0, is symmetric around 0, and single-peaked at 0. Denote the distribution of posteriors q_t given a prior q_{t-1} and policy x_t as $\tau(q_t|q_{t-1}, x_t)$. Suppose x'_t is farther from x_{LT} than some other policy x_t . The variance of posteriors of $\tau(q_t|q_{t-1}, x'_t)$ is larger than $\tau(q_t|q_{t-1}, x_t)$ but has the same mean; policies farther from x_{LT} reveal more information.

The leader’s problem can now be expressed as:

$$V(q) = \max_{x \in \text{NR}(q)} \underbrace{u_\ell(x)}_{\text{Flow utility}} + \delta \underbrace{\int V(q') \tau(q'|q, x)}_{\text{Value of information}} \quad (8)$$

The value of information is $V(q)$ integrated over the potential posteriors $\tau(q'|q, x)$. When $x = x_{LT}$, $\tau(q'|q, x)$ places mass 1 on the prior q ; as x grows more extreme, the variance of posteriors increases. There is hence an aversion to information if $V(q)$ is concave, which holds if the graph of $\text{NR}(q)$ is convex; this is true in the baseline model (Figure 2), but may not hold generally (e.g. Figure 4). The following proposition characterizes $x_\ell(q)$.

Proposition 4. *Suppose the graph of $\text{NR}(q)$ is convex. There exists a threshold $\underline{q} \in (0, 1]$ such that:*

1. *if $q \leq \underline{q}$, $x_\ell(q) = \bar{x}(q)$. The leader plays the most extreme policy preventing overthrow.*
2. *If $q \geq \underline{q}$, the leader implements $x_\ell(q) \in [\min\{x_{LT}, \underline{x}(q)\}, \bar{x}(q)]$. Moreover, for δ high, $x_\ell(q) < \bar{x}(q)$; i.e., she moderates her actions to pursue the learning trap.*

Multiple States of the World The benchmark model considers two states of the world. However, suppose there are three states: a “pro-leader,” “anti-leader,” and “middle” state, where the optimal policy for constituents is moderate. The “Three States of the World” part of the Online “Additional Results” shows that the model’s intuitions generalize to this setting.

While leaders cannot totally shut down information between all three states, they can pool information “pairwise.” For example, they can implement a policy that separates the “anti-leader” state but keeps constituents uncertain between the “pro-leader” and “middle” states. Agents will eventually learn that either the “anti-leader” state is true, or that one of the “pro-leader” and “middle” states is true, but not which. Lemma 9 shows that at intermediate beliefs, the leader pools the “anti-leader” with either the “pro-leader” or “middle” state, but never separates the “anti-leader” state; in either case, the leader moderates her actions instead of pursuing her preferred policy.

The model with two states can represent the terminal history of a model with more states. Suppose the leader pools the “anti-leader” and “pro-leader” states while separating the “middle” state. Then, either the “middle” state realizes or constituents learn that the “middle” state does not hold — in which case the model returns to the benchmark with a pro and anti-leader state.

Model with Ambiguity Aversion We have assumed all agents have a common prior q in the pro-leader state. However, the people may be diverse in their beliefs; or, the leader may be uncertain about where in a larger set constituents' subjective beliefs lie. How does this affect leaders' behavior?

We thus study the leader's behavior under *ambiguity aversion*: instead of behaving as if the people's belief in the pro-leader state is q , she believes their beliefs lie in $\mathcal{Q} \subseteq [0, 1]$. The leader behaves as a "maxmin expected utility" maximizer in the sense of Gilboa and Schmeidler (1989), meaning she chooses each policy anticipating the corresponding "worst case scenario." This behavior can also reflect the leader's aggregation of a diverse set of beliefs that requires appeasing the "most pessimistic" of her constituents.

We start with an example evaluating the leader's continuation utility. Suppose the leader can pursue either $x = 0$ or $x = 1$ today and avert overthrow. The prior q moves to the posterior q' , which is always either 0, 1, or q . The leader evaluates the problem as:

$$\max_{x \in \{0,1\}} \min_{q' \in \mathcal{Q} \cup \{0,1\}} u_\ell(x) + \delta V(q') = \max_{x \in [0,1]} u_\ell(x) + \delta \underline{u} \quad (9)$$

She thus evaluates information revelation as if the anti-leader state materializes. Returning to the original problem, suppose the leader believes q lies in a compact set $\mathcal{Q} \subseteq [0, 1]$ such that $\mathcal{Q} \cap (0, 1) \neq \emptyset$. Her problem is:

$$V(\mathcal{Q}) = \max_{x \in [0,1]} \min_{q \in \mathcal{Q}} \begin{cases} u_\ell(x) + \delta \min_{q' \in \mathcal{Q}} V(\mathcal{Q}) & x \in \text{NR}(q), x = x_{LT} \\ u_\ell(x) + \delta \min_{q' \in \mathcal{Q} \cup \{0,1\}} V(q') & x \in \text{NR}(q), x \neq x_{LT} \\ -L & x \notin \text{NR}(q) \end{cases} \quad (10)$$

If she pursues x_{LT} , no information is revealed; her continuation utility is equal to $\min_{q' \in \mathcal{Q}} V(\mathcal{Q})$; because $x_{LT} \in \text{NR}(q)$ for all $q \in \mathcal{Q}$, if x_{LT} is implementable today, it is also implementable tomorrow. If she implements $x \neq x_{LT}$, information is revealed, which she evaluates as in (9). There may also be no policies in $\text{NR}(q)$ for all q , in which case the leader may be deposed.

The following result shows that, where possible, the leader will implement the learning trap, since she evaluates information as if it favors the anti-leader state. If *no policies* surely avert over-

throw, the leader pursues $x = 1$ and hopes for the best.

Proposition 5. *Let $\mathcal{Q} \subseteq [0, 1]$ be a compact set of constituent beliefs such that $\mathcal{Q} \cap (0, 1) \neq \emptyset$. Suppose the leader operates as a maxmin expected utility maximizer and δ is large. Let $x_\ell(\mathcal{Q})$ be the leader's optimal policy conditional on believing the people's belief q lies in $\mathcal{Q}$. Then:*

- *if $x_{LT} \in \text{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q}) = x_{LT}$;*
- *if there exists x such that $x \in \text{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q})$ is the highest policy that prevents overthrow;*
- *if there does not exist $x \in [0, 1]$ such that $x \in \text{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q}) = 1$.*

4 Application: The 1861 Reforms as a Learning Trap in Imperial Russia

This section illustrates how the model's informational insights can help understand long-lived regimes' policy choices. I study the reforms ending serfdom in Imperial Russia, showing how the Tsar's pursuit of a "middle-ground" economic system that inefficiently mixed communal decision-making with elements of a market economy resembled a "learning trap." I map the Russian case to the model and argue that the choice of this "learning trap" was a response to uncertainty about the alignment of the Tsar's preferred policy with Russians' interests. Comments on alternative equilibria of the model and explanations for the reforms are expounded in the Online "Additional Case Study Material."

4.1 Background

Before 1861, most Russian peasants provided services or payments to the state or a noble in exchange for use of arable land. Peasants working noble land were "serfs," and the system closely resembled slavery.¹² Seigneurs oversaw a "commune" of households in charge of repartitioning land, solving disputes, and managing resources. Many peasants possessed limited control over production decisions, had few rights, and could be tied to land in the event of sale (Vasudevan (1988, 209), Nafziger (2010, 382), Nafziger (2016, 775-776) Dennison (2014, 252-257)). After

¹²Peasants rarely owned land, but both before and after 1861 owned small pieces of private property. There was heterogeneity in this arrangement; Nafziger (2012) shows how peasants on State-owned lands possessed some more individual property rights than serfs.

Russia's defeat in the 1853 Crimean War, discourse about alternatives to Serfdom began circulating (Starr, 2015, 57). Many educated Russians maintained that the Tsar lost the war due to economic backwardness. Some nobles argued that dissolving communes in favor of (serf-owned) private property would improve economic growth (Starr, 2015, 53-63), while others favored systems where nobles would continue to own land but peasants could rent and work it "on the basis of free competition" (Khristoforov, 2009, 65).

Regime's Preferred Policy The Russian regime, headed by Tsar Alexander II, likely preferred a liberal path to economic modernization that resembled Western Europe — characterized by household private property, individual rights to output, and freedom for labor to move to cities — due to its implications for state development, fiscal capacity, and the military. Russia's widening military gap with its neighbors was attributed to serfdom's role in stifling the emergence of industry (Zenkovsky (1961, 284), Emmons (1966, 48-50)) and the State's ability to draw army reserves from the peasantry (Moon, 2014, 53-54). Endowing peasants with private property could allow labor to flow to the cities — generating industrial growth — and mitigate frictions that made it difficult to recruit peasants from nobles' land. A liberal system could also accelerate the development of state institutions through its fiscal effects (Dennison, 2023, 642). Under serfdom, tax collection was inefficiently outsourced to nobles who siphoned surpluses (Dennison, 2020, 2023). Liberal reform could weaken the nobility's bite, increasing revenue for the State (Moon, 2014, 33).¹³ Reformers hoped to use fiscal revenues to increase state capacity and fund the expansion of corporate forms and military ventures (Dennison, 2023, 642). In an ideal world, liberal reform would be *aligned* with constituents' interests as well: it could increase worker mobility and productivity, generate economic growth via industry, enhance Russia's ability to defend itself during war, and increase quality of life for rural households (Pereira, 1980, 108).

Model Mapping We view the Russian setting through our model as follows. One extreme policy is full-on liberal reform, while the other is the status quo of serfdom. In the "pro-leader" state of the world, the government's preferred policy — liberal reform — would be aligned with the people's economic interests — a combination of the peasantry and nobility.

¹³If liberal reform could *substantially* increase surpluses, the government would be in a better place, even if it had to compensate elites for dispossessing their land.

In the anti-leader state of the world, the government's interests would be unaligned with its constituents', who would force the government to maintain serfdom or pursue non-liberal approaches to development. Accordingly, many Russians did believe that elements of serfdom — particularly the commune — offered benefits over liberal reform. Slavophilic ideologues romanticized the peasant as a self-sufficient "bearer of his own culture" (Khristoforov, 2009, 62) and argued the commune should be left untouched. Some believed private property could lead to a countryside marred with disorder, requiring seigniorial intervention to prevent exploitation of commons (Pravilova, 2014). Others contended that peasants were not aware of issues with the pre-1861 system (Field, 1976, 53), even arguing peasants preferred communal land (Kingston-Mann, 1991, 34). A minister of internal affairs contrasted Russia's "uninterrupted tranquility" under serfdom with the "[i]nternal discord and revolts" plaguing Europe's experiments with liberal reform (Polunov et al., 2015, 87). The Russian countryside was not beset by an endemic economic crisis (Moon, 2014, 19-22), and nothing had "challenged the basic structure of local institutions" (Starr, 2015, 52).

At the eve of the 1860s, these debates imply that there was *uncertainty* over whether the government's preferred policy of liberal reform was aligned with Russians' interests. The "beginning" of the model can be thought of as the ascension of Alexander II to power in 1855 and a "shock to beliefs" triggered by the Crimean War that shifted public opinion towards the pro-leader state. Relatedly, if we interpret the pro and anti-leader states as representing constituents' subjective models of the world, we can view the Crimean War as triggering the "introduction" of a pro-leader/liberal reform model.

4.2 Learning Trap, 1861-1905

Government's Fears of Information Revelation How did the government respond to uncertainty about whether liberal reform would be aligned with its constituents' interests? The Tsar understood that its *policy* choices could reveal *information* about the effects of policies that could potentially *constrain* it in the future, particularly if liberal reform backfired. Because Alexander II's government worried liberal policies could fail to *sufficiently* grow the Russian economy if the pro-leader state were not true, it viewed "communal land tenure and mobility restrictions [as] nec-

essary precautions” in the face of this uncertainty (Dennison, 2014, 62). Endowing peasants with private property *could* lead to excessive alienation of rural land, limiting the government’s ability to collect revenue, stressing state capacity (Dennison, 2023, 644). External validity arguments that Westernization was inappropriate for Russia accompanied fears of “the rise of an industrial proletariat, ruthless competition, unemployment, and other evils of industrial capitalism” (Polunov et al., 2015, 61). One of Alexander II’s committees thus concluded that “the Order of the State might be shaken” by an emancipation that did *not* tie peasants to land (Field, 1976, 75), suggesting that mixing some elements of communalism into the ensuing reforms could potentially tide these *informational fears* that would materialize if full-on liberal reform were to backfire.

Terms of Emancipation Thus, while the Tsar’s emancipation reforms beginning in 1861 freed peasants from the coercive status quo of seigneur-peasant agriculture and broadened civil rights (Dennison, 2014), they did *not* remove the peasant commune and counterintuitively used it to constrain household labor mobility and capacity to acquire private property.

Nobles were compensated by the State for relinquishing land that was redistributed between members of peasant communes, financed by mortgage-like redemption made by serfs to the State over 49 years. After payments were made, ex-serfs would privately own their allotted land but were otherwise *communally* responsible for redemption and tax payments. Households could not legally sell, lease, transfer, alienate, or use land as collateral *until* redemption payments were complete, restricting freedom to abandon agriculture (Dower and Markevich (2019, 234), Nafziger (2012, 5-6), Nafziger (2016, 776), (Dennison, 2014, 259)).

Intermediate Policy The 1861 reforms mixed communal serfdom with liberal institutions. An external seigneur no longer regulated economic decisions. Peasants could privatize communal land by paying off redemption obligations (Nafziger, 2010, 383). Peasants owned houses, gardens, and rights to output (Kingston-Mann, 1991, 35). Communes (and sometimes households) even engaged in market land transactions with formal credit backing (Nafziger, 2010, 384). However, the persistence of the commune and its codification by the State clashed with individual economic incentives. Land allotments, repartitions, and redemption obligations were enacted communally, sometimes against households’ wills. Peasants owned scattered strips of land, at best

allowing for diversification, at worst increasing travel times and disincentivizing technological experimentation, slowing agricultural productivity growth (Williams (2013, 52-53, 65-66), Dower and Markevich (2019, 243)).

Mobility Restrictions The commune restricted labor mobility and supply by controlling the issuance of documents to pursue outside work (Nafziger, 2012, 7). Because households were communally responsible for redemption payments, those with larger obligations were incentivized to enforce mobility restrictions and free-ride on others. Nafziger (2010) notes: “Income-maximizing peasant households may have been unable to freely allocate resources between agricultural production and non-agricultural activities if they were subject to collective decision-making. . . By potentially inhibiting land transfers. . . and restricting the allocation of labor outside the village, the communal system may have introduced wedges between the shadow and market values for these factors of production” (384). Notably, these frictions stifled the flow of labor to cities, dampening the emergence of Russia’s industrial sector and contradicting one of the original motives for reform.¹⁴

Implications Many contemporaries felt the Tsar’s reforms did not go far enough. In practice, “[f]ew surpluses could be wrung from peasants who were already obligated to repay the nobility for their freedom and their land and who continued to face obstacles to the most efficient allocation of their resources” (Dennison, 2023, 644-5). These realities make the choice to maintain the commune odd; if liberal reform could certainly result in large economic gains, the State would have been better off fiscally than in a middle ground where both peasant surplus and the ability to pay the nobility seemed constrained.

Instead, fears that information revelation could strain the government’s coffers and reputation were it to backfire — revealing that its preferences were unaligned with the people — provide an informational explanation for this middle-ground. From early on, the Tsar’s government explicitly believed that a system that ending many of serfdom’s abuses but still tying peasants to agricultural land could mitigate these informational threats.

Indeed, over the decades, elites’ and the State’s inability to conclude whether the commune

¹⁴Indeed, Nafziger (2010) argues that weaker communal structures around Moscow allowed it to develop a stronger industrial sector.

would have been better than individualized agriculture suggests that the post-1861 middle-ground halted learning about the effects of alternative policies. Despite the commune's apparent discouragement of technological adoption, Kingston-Mann (1991) documents that when innovation did occur, it *spread* more quickly in districts with greater communal intensity. While many in the government still believed in the pursuit of a liberal, Western path, one source cites a government official, "an enthusiastic critic of the commune" (Kingston-Mann, 1991, 49), who conceded that the commune's ability to provide mutual aid during emergencies was a benefit that could not be realized under private property. Decades later, officials contended that limited cases where peasants were permitted to sell land caused exploitation of peasant decision making (Wcislo, 2014, 89), casting doubt about the appropriateness of markets that would accompany liberal reform.

The *general equilibrium* effects of reform likely differed from the partial equilibrium experience of individual villages toying with private incentives. Regional heterogeneity in the enforcement of communal institutions allowed some households to observe the effects of private incentives, accumulating wealth by participating in rental markets and non-agricultural labor (Nafziger, 2010, 385). Nevertheless, the contemporary remarks above suggest high-level uncertainty about whether liberal reform was surely better, even decades after 1861.

Not only did the Tsar's government hold power for decades, but except for a spike in disturbances right after 1861, reported peasant discontent remained low in the following years (Finkel et al., 2015, 1000). In this sense, like a learning trap, the Tsar's intermediate policy was successful in tiding threats of overthrow. However, beyond stifling Russia's industrial sector and potentially slowing agricultural adoption in the medium term, Markevich and Zhuravskaya (2018) go as far as to argue that the allocation of property rights to communes instead of households dampened Russian economic growth into the 20th century.¹⁵

4.3 Other Equilibria and Interpretations for Learning Trap

While the model predicts that the Tsar would pursue a learning trap at intermediate beliefs in the pro-leader state, it also predicts that at extreme beliefs, the government would experiment with

¹⁵Some historians point to redemption payments as the first order inefficiency impeding livelihood in post-1861 ex-serf communities. Interestingly, Gerschenkron (1962) suggests that the commune *worsened* the bite of redemption payments. Households with particularly large debt obligations would have an incentive to free-ride on other members of the commune, affecting commune enforcement of labor mobility and production incentives which undoubtedly affected households' very ability to make payments.

liberal reform. The first part of the “Additional Case Study Material” argues that this was the case: after strong shocks to beliefs in 1905 favoring liberal reform, the government experimented with private property under Pyotr Stolypin.

Second, the historical literature centers two explanations for the Tsar’s middle ground: bargaining with the nobility and low state capacity. The second part of the “Additional Case Study Material” argues that while these factors undoubtedly shaped the 1861 reforms, the diminishing power of the elites, similarities in the experiences of ex-serfs to peasants on state-owned lands, and lack of policy variation over decades cast doubt on whether these forces alone explain the intermediacy of the emancipation reforms. However, viewing the reforms as a *learning trap* resolves these puzzles via a complementary informational mechanism.

5 Application: Anticipated Information Revelation and Political Contagion

Can the model’s insights explain why certain regions of the world experience more policy experimentation? To answer this question, I extend the baseline model to allow for the leader and people to also learn about the state of the world from external sources. I show that external information revelation of this sort increases leaders’ willingness to experiment with their preferred policies from both a direct channel (exogenous information) and an indirect channel (the leader herself is more likely to prefer experimentation, increasing learning).

I show how this insight allows us to understand the interplay of policy contagion — where information about the effects of policies spreads from one polity to another — and political fragmentation. Specifically, I interpret 19th century Europe as a region where the probability of external information revelation was high and China as a region where it was low, and argue that the model can explain why European nations experimented with indirect rule — experiencing high political turnover in the process — while China did not.

5.1 Information Revelation

Information revelation may be *anticipated* when leaders are consciously aware of external sources of inference. Officials in regions plagued by earthquakes or hurricanes may foresee that these

disasters will eventually occur and reveal information about disaster relief policies. Another application is policy contagion, where a group of countries with similar institutions can learn from their policy experiences. Suppose that all these countries are stuck in a learning trap except for one, which commits to implement informative policies. Neighboring countries use these informative policies as an additional source of policy inference, meaning their leaders now face an additional threat of information revelation from external sources.

I model anticipated information shocks as follows. Let $\eta \in (0, 1)$ be small. With probability η at the end of every period, the true state of the world is exogenously revealed. The leader's problem, anticipating the exogenous revelation of information, can be written as:

$$\max_{x \in \text{NR}(q)} \frac{2\sigma(1-\delta)u_\ell(x) + \delta((1-\eta)\Delta(x) + 2\sigma\eta)\Psi(q)}{2\sigma(1-\delta) + \delta((1-\eta)\Delta(x) + 2\sigma\eta)} \quad (11)$$

Note that when $x = x_{LT}$, the leader's value is now

$$\frac{(1-\delta)x_{LT} + \delta\eta\Psi(q)}{(1-\delta) + \delta\eta}$$

Implementing x_{LT} is no longer a “safe option” for the leader that shuts down learning. Anticipating that she can no longer *fully* control information, she may take extreme actions to enjoy more flow utility. This intuition is reflected in the following proposition.

Proposition 6. *Suppose η increases. Then, $\underline{q}$ decreases and $\bar{q}$ increases. The leader plays the learning trap for fewer values of q .*

Anticipated information revelation improves policy learning compared to the unanticipated or baseline model. While there is mechanically more information revelation due to the realization of shocks, because $\underline{q}$ decreases, the leader experiments with her preferred policy for a larger range of q values, meaning learning also increases through an *endogenous* channel. As learning increases, q is more likely to move to 0. This means that fragmented regions plagued by policy contagion will see more governments conceding to their people — e.g. movements from absolute to constitutional monarchy — or, insofar as lower x is also a shorthand for increased external threats of overthrow, leaders' terms will be shorter, corresponding to more frequent political turnover.

5.2 Political Fragmentation in Europe and China

The model's prediction — politically fragmented regions of the world exhibit larger policy variation, more policy experimentation, and more frequent political turnover — is consistent with stylized historical facts differentiating Western Europe and China. It explains how the spread of constitutional monarchies and representative democracies in fragmented Western Europe contrasted with the consistent stability of consolidated Imperial China. Because systems of indirect rule were not widely experimented with by Eurasian states until the 1700s, we view our “uncertain policy” as the optimal degree of democracy or autocracy. The policy space corresponds to direct rule (e.g. monarchy) on one end and indirect rule (e.g. republican democracy) on the other end. Constitutional monarchies can be thought of as situations where autocrats begrudgingly cede power to the people but remain in office.

European countries around 1800 had a high η ; Europe was populated by dozens of sovereign states — on average 85 between the years 1000 and 1799 (Dincecco and Wang, 2018)) — and thus possessed rich sources of policy information. England's “Glorious Revolution” at the end of the 17th century marked Europe's first constitutional monarchy (Kurian, 2011) and one of the first sources of data on indirect rule. After the French Revolution, rulers began to actively worry about citizens' desires for indirect rule (Evans and Von Strandmann, 2002, 10) as demands for freedom of speech, press, and disposal of private property became more prominent (Sperber, 2005, 66-67). By the first half of the 19th century, both Britain (constitutional monarchy) and America (representative democracy) were providing information on the efficacy of indirect rule. As these informational pressures strengthened, by 1850, most European states shifted to some form of government partially controlled by an elected legislature (Tilly, 1989). Some governments peacefully ceded power, while others had power wrested from them as part of the “Age of Revolutions.” A threat of external information revelation forced governments to experiment with indirect rule, increased learning via convergence to indirect rule, and led to political upheaval.

In contrast, China was ruled by a single, unified State uninterrupted for a millennium. No states shared China's scale or political structure, suggesting it had low η in 1800. China possessed relatively integrated economic institutions that allowed its central government to oversee trade and a tightly-managed bureaucracy composed of strong elite networks (Rosenthal and Wong,

2011). The state's Confucian ideology centered political continuity and stability, contrasting with the “disruptive progress” of Western Europe. The entrenched power of the emperor coupled with a shared ideology of “stability” among the state and elites arguably prevented the emergence of ideologies promoting indirect rule, which was only strengthened by the lack of external information (Mokyr, 2016, ch. 16). China's high (although stagnant) economic standards in the late 1700s evidence a degree of policy *moderation* that was not seen in Western Europe (Pomeranz, 2021). Rare threats of overthrow came from regional elites and seemingly strengthened — rather than eroded — the collective power of the emperor and elites (Dincecco and Wang, 2018).

The “high η ” environment of Western Europe in the 18th and 19th centuries was accompanied by greater policy variation, concessions to liberal demands, and political transitions, in contrast to the consolidated “low η ” environment of China, which experienced little variation and minimal turnover. This prediction is consistent with Mokyr (2016), who argues that “there were repressive and reactionary regimes galore in Europe, but the interstate competitiveness constrained their ability to enforce a specific orthodoxy” (317). This prediction also complements a literature arguing that Europe's “political fragmentation” drove the Great Divergence by fostering competition over technology (Diamond, 2005), intellectual networks (Mokyr, 2016), and institutional innovation (Cox, 2017).

6 Additional Applications

Finally, I provide additional examples of how learning traps can help understand leaders' policy decisions. Take Soviet Premier Leonid Brezhnev's “developed socialism,” marketed as a transition phase between a state's development of socialist infrastructure and convergence to a communist society (Sandle, 2002, pp. 165-166). Despite his predecessor Khrushchev's commitment to achieving full communism by 1980, Brezhnev was notorious for refusing to adopt the necessary reforms to achieve this goal, pursuing counter-reform to his own 1965 planning and production policies, and making it hard to argue that growth would have been higher *or* lower without the counter-reforms (Harrison, 2002, pp. 57-58). The model can provides insights into Brezhnev's dilemma: he was unsure whether communism would work — hence whether his preferred policy was aligned with constituents' — and pushed an inefficient middle ground that prevented constituents from

learning the effects of “true communism” but sustained Soviet power.

A second example comes in the nebulous attitude of Franklin D. Roosevelt towards health-care reform. FDR refused to include national health insurance in the 1935 Social Security Act, instead dropping it “at the last minute” (Quadagno, 2006, p.23). While the President convened a committee on medical care that recommended the expansion of public health programs and convened a conference to guide the design of those programs, he “asked conference organizers to be circumspect about any possible next steps” (Hoffman, 2012, p. 29), refused to endorse a Senate call to fund state-level insurance, and permitted the proliferation of private insurance. Despite the opposition of the physician lobby, FDR had the power to expand government intervention in healthcare, and his actions are inconsistent with his belief in healthcare as a human right (Hoffman, 2012, p. 36). One way to read FDR’s actions is that he believed the public was uncertain whether government-run health programs would actually be aligned with their broad interests, and that a failure of these programs could curtail future expansions of the welfare state.

Finally, learning traps can also be applied to states’ willingness to wage “limited” rather than “total” war. The United States’ extended involvement in Afghanistan in the 2000s and 2010s was characterized by deliberate limitations in troop deployment that hampered efforts “to gain sufficient military victories to force a political settlement” Griffin (2014). This limited involvement accompanied widespread public uncertainty about whether there was any hope of salvaging Afghan political stability through external military intervention (Biddle et al., 2010). Viewed through the model, a leader who is uncertain about her country’s ability to win a war may moderate her actions and pursue “limited war.” Intuitively, full-scale involvement reveals surely whether victory is possible — which can be risky for leaders if victory is impossible. By contrast, limited war provides cover: any gains or losses can be attributed to partial involvement. Yet, just as learning traps must be perpetually maintained to preserve uncertainty, a leader may need to sustain a limited war over an extended period to uphold these reputational effects.

7 Conclusion

This paper studies a repeated model where a leader is uncertain about whether her preferred policy benefits her constituents. Her constituents learn about the effects of policies from the leader’s

policy choices. Thus, the leader's choices generate *information* that affects what she can implement tomorrow. This informational externality generates a *nonmonotonicity* between a leader's willingness to experiment with her preferred policy and constituents' beliefs in whether their preferences are aligned with the leader's. When beliefs are intermediate — so the optimal policy for constituents is unknown — the leader pursues a spatially intermediate “learning trap policy” — shutting down information revelation when it would be most useful. She pursues extreme policies only at extremal beliefs, when constituents are relatively certain that the optimal policy is (or is not) the leader's preferred policy.

These predictions provide insight into the policy choices of long-lived leaders in persistent dictatorships or entrenched incumbencies — who halt learning when the effects of policies are unknown, but may experiment if policies are sufficiently likely to benefit their constituents in expectation. These predictions contrast with the behavior of short-lived leaders who “gamble for resurrection,” only experimenting when they are *unlikely* to be aligned with their constituents. I show how the insights from the repeated model are robust to information asymmetries, additional preferences for constituent learning, multiple states of the world, uncertainty about constituents' beliefs, and external information revelation.

I then use a case study of the end of Serfdom in Russia to illustrate how the informational mechanisms of the model plausibly played a role in the policy choices of Tsar Alexander II. In the late 1850s, the Tsar wanted to modernize the Russian economy by pursuing Western European style economic reform. However, his government was unsure whether this preferred policy of “liberal reform” would actually work in Russia due to its cultural and institutional differences with the West. As a result, the government implemented an intermediate policy that mixed elements of communal and liberal institutions, creating an inefficient “middle ground” that distorted labor and production decisions, restricting economic, fiscal, and industrial growth. However, I argue that the choice of this middle ground can be thought of as a learning trap: by preventing constituents from being able to discern the effects of different policies, the government was able to halt learning and keep itself in power for decades.

I then augment the model with external information revelation to study how policy contagion can affect leaders' actions. Allowing leaders and constituents to learn about the effects of different policies from external sources shrinks the range of beliefs for which leaders want to pursue learn-

ing traps, since leaders cannot fully control information revelation. By interpreting this exogenous information as the result of policy experimentation in an externally valid country, I show how the model can explain why Europe experienced a great deal of political upheaval and convergence to systems of indirect rule in the 19th century, while China did not, complementing “fragmentation” hypotheses of the Great Divergence. Furthermore, I show how learning traps can also be applied to understand Brezhnev’s refusal to adopt “full communism” in the USSR, FDR’s unwillingness to include healthcare reform as part of the New Deal, and why polities like the US may wage limited wars.

Ultimately, long-lived leaders pursue learning traps precisely when their constituents are uncertain about the optimal policy for them, and learning would be most useful. While endowing leaders with term limits may provide an appealing solution to this dilemma, the “gambling for resurrection” literature suggests that this will only improve policy experimentation if a leader’s preferred policy is *misaligned* in expectation with her constituents, and may in fact encourage learning traps if preferences are *aligned*. Thus, future work will further investigate accountability mechanisms that encourage efficient policy experimentation in both repeated and term-limited settings.

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Proofs of Results with Patient People

Proof of Solution to People's Repeated Problem

The following proposition characterizes the solution to the people's problem when there is no preference for learning. We normalize flow utility to $(1-\delta)(\mathbb{E}[y|x] - \kappa|x-x'|)$ and define $k = \kappa(1-\delta)$. The people's Bellman equation is $W(q, x')$ and its solution $x_p(q, x')$:

$$W(q, x') = \max_{x \in [0,1]} (1-\delta)\mathbb{E}[y|x] - k|x-x'| + \delta \left(\frac{\Delta(x)}{2\sigma} (qW(1, x) + (1-q)W(0, x)) + \left(1 - \frac{\Delta(x)}{2\sigma}\right) W(q, x) \right).$$

Proposition 3. $x_p(q, x')$ is also the solution to the following equation:

$$\max_{x \in [0,1]} \frac{2\sigma(1-\delta)\mathbb{E}[y|x] + \delta\Delta(x)\Phi(q, x)}{2\sigma(1-\delta) + \delta\Delta(x)} - k|x-x'|$$

where $\Phi(q, x) = qW(1, x) + (1-q)W(0, x)$. Moreover, if $k < f'_g(x_{LT}) < 2k$, there exists c and a nonempty set $LT \subseteq [0, 1] \times X$ with $LT \ni (1/2, x_{LT})$ such that for all $(q, x') \in LT$, $W(q, x') \leq y_{LT} + c \leq W(0, x'), W(1, x')$.

The proof for this result is contained in Lemmata 1-5.

Lemma 1. Suppose x^* is a solution to the functional equation defining $W(q, x')$. Then, x^* is also a solution to the functional equation defining $W(q, x^*(q, x'))$.

Proof. Suppose by contradiction that x^{**} solves the functional equation associated with $W(q, x^*)$ but not $W(q, x')$. Since both x^* and x^{**} are always feasible, this implies

$$\begin{aligned} & (1-\delta)\mathbb{E}[y|x^{**}] - k|x^{**} - x^*| + \frac{\delta\Delta(x^{**})}{2\sigma} (qW(1, x^{**}) + (1-q)W(0, x^{**})) + \delta \left(1 - \frac{\Delta(x^{**})}{2\sigma}\right) W(q, x^{**}) \\ & > (1-\delta)\mathbb{E}[y|x^*] + \frac{\delta\Delta(x^*)}{2\sigma} (qW(1, x^*) + (1-q)W(0, x^*)) + \delta \left(1 - \frac{\Delta(x^*)}{2\sigma}\right) W(q, x^*) \end{aligned}$$

which also implies

$$\begin{aligned}
& (1-\delta)\mathbb{E}[y|x^{**}] - k|x^{**} - x^*| - k|x^* - x'| + \frac{\delta\Delta(x^{**})}{2\sigma}(qW(1, x^{**}) + (1-q)W(0, x^{**})) \\
& + \delta(1 - \frac{\Delta(x^{**})}{2\sigma})W(q, x^{**}) \\
& > (1-\delta)\mathbb{E}[y|x^*] - k|x^* - x'| + \frac{\delta\Delta(x^*)}{2\sigma}(qW(1, x^*) + (1-q)W(0, x^*)) + \delta(1 - \frac{\Delta(x^*)}{2\sigma})W(q, x^*)
\end{aligned}$$

However, since $k|x^{**} - x'| \leq k|x^{**} - x^*| + k|x^* - x'|$, this implies

$$\begin{aligned}
& (1-\delta)\mathbb{E}[y|x^{**}] - k|x^{**} - x'| + \frac{\delta\Delta(x^{**})}{2\sigma}(qW(1, x^{**}) + (1-q)W(0, x^{**})) + \delta(1 - \frac{\Delta(x^{**})}{2\sigma})W(q, x^{**}) \\
& \geq (1-\delta)\mathbb{E}[y|x^{**}] - k|x^{**} - x^*| - k|x^* - x'| + \frac{\delta\Delta(x^{**})}{2\sigma}(qW(1, x^{**}) + (1-q)W(0, x^{**})) \\
& + \delta(1 - \frac{\Delta(x^{**})}{2\sigma})W(q, x^{**}) \\
& > (1-\delta)\mathbb{E}[y|x^*] - k|x^* - x'| + \frac{\delta\Delta(x^*)}{2\sigma}(qW(1, x^*) + (1-q)W(0, x^*)) + \delta(1 - \frac{\Delta(x^*)}{2\sigma})W(q, x^*)
\end{aligned}$$

contradicting that x^* solved the functional equation associated with $W(q, x')$. $\square$

Lemma 2. *There exist $0 \leq \underline{x} < x_{LT} < \bar{x} < 1$ such that*

$$W(1, x') = \begin{cases} f_g(\bar{x}) - k(\bar{x} - x') & x' \leq \bar{x} \\ f_g(x') & x' > \bar{x} \end{cases} \quad W(0, x') = \begin{cases} f_b(\underline{x}) - k(x' - \underline{x}) & x' \geq \underline{x} \\ f_b(x') & x' < \underline{x} \end{cases}$$

Proof. Suppose $q = 1$ and $x' \leq 1$. $W(1, x') = \max_{x \in [0,1]} (1-\delta)f_g(x) + \delta W(1, x) - k|x - x'|$. By Lemma 1, this simplifies to $\max_{x \in [0,1]} f_g(x) - k|x - x'|$, which is solved by $\bar{x} = 1 - k/2$. That these points are to the right of x_{LT} follows from $f'_g(x_{LT}) > k$. A similar argument applies for $W(0, x')$. $\square$

Lemma 3. *Suppose x^* solves the functional equation defining $W(q, x')$ and $q \in (0, 1)$. Then:*

$$W(q, x^*) = \frac{2\sigma(1-\delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q, x^*)}{2\sigma(1-\delta) + \delta\Delta(x^*)}$$

where $\Phi(q, x) = qW(1, x) + (1-q)W(0, x)$.

Moreover, $W(q, x') = \max_x \frac{2\sigma(1-\delta)\mathbb{E}[y|x] + \delta\Delta(x^*)\Phi(q, x)}{2\sigma(1-\delta) + \delta\Delta(x)} - k|x - x'|$.

Proof. Since x^* solves the functional equation associated with $W(q, x')$, by the previous lemmata

it also solves the one associated with $W(q, x^*)$. Hence:

$$\begin{aligned} W(q, x^*) &= (1 - \delta)\mathbb{E}[y|x^*] + \frac{\delta\Delta(x^*)}{2\sigma}\Phi(q, x^*) + \delta\left(1 - \frac{\Delta(x^*)}{2\sigma}\right)W(q, x^*) \\ &= \frac{(1 - \delta)\mathbb{E}[y|x^*] + \frac{\delta\Delta(x^*)}{2\sigma}\Phi(q, x^*)}{1 - \delta\left(1 - \frac{\Delta(x^*)}{2\sigma}\right)} = \frac{2\sigma(1 - \delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q, x^*)}{2\sigma(1 - \delta) + \delta\Delta(x^*)} \end{aligned}$$

$W(q, x')$ can then be written as:

$$\begin{aligned} W(q, x') &= \max_{x^*} (1 - \delta)\mathbb{E}[y|x^*] + \frac{\delta\Delta(x^*)}{2\sigma}\Phi(q, x) \\ &\quad + \delta\left(1 - \frac{\Delta(x^*)}{2\sigma}\right) \frac{2\sigma(1 - \delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q, x)}{2\sigma(1 - \delta) + \delta\Delta(x)} - k|x^* - x'| \\ &= \max_x \mathbb{E}[y|x] \left((1 - \delta) + \delta\left(1 - \frac{\Delta(x)}{2\sigma}\right) \frac{2\sigma(1 - \delta)}{2\sigma(1 - \delta) + \delta\Delta(x)} \right) \\ &\quad + \delta\Phi(q, x) \left(\frac{\Delta(x)}{2\sigma} + \left(1 - \frac{\Delta(x)}{2\sigma}\right) \frac{\delta\Delta(x)}{2\sigma(1 - \delta) + \delta\Delta(x)} \right) - k|x - x'| \\ &= \max_x \mathbb{E}[y|x] \left((1 - \delta) \left(1 + \frac{2\sigma\delta - \delta\Delta(x)}{2\sigma(1 - \delta) + \delta\Delta(x)}\right) \right) \\ &\quad + \delta \frac{\Delta(x)}{2\sigma} \Phi(q, x) \left(1 + \frac{2\sigma\delta - \delta\Delta(x)}{2\sigma(1 - \delta) + \delta\Delta(x)}\right) - k|x - x'| \\ &= \max_x \frac{2\sigma(1 - \delta)\mathbb{E}[y|x] + \delta\Delta(x)\Phi(q, x)}{2\sigma(1 - \delta) + \delta\Delta(x)} - k|x - x'| \end{aligned}$$

□

Lemma 4. Suppose $x' \in [\underline{x}, \bar{x}]$. Then $x_p(q, x') \in [\underline{x}, \bar{x}]$.

Proof. Note that our problem can be written as:

$$\begin{aligned} W(q, x') &= \max_{x \in [0, 1]} \frac{2\sigma(1 - \delta)}{2\sigma(1 - \delta) + \delta\Delta(x)} (\mathbb{E}[y|x] - k|x - x'|) \\ &\quad + \frac{\delta\Delta(x)}{2\sigma(1 - \delta) + \delta\Delta(x)} (\Phi(q, x) - k|x - x'|) \end{aligned}$$

First, since $\frac{\partial^2}{\partial q \partial x} \mathbb{E}[y|x] = f'_g(x) - f'_b(x) > 0$, since $\arg \max f_g(x) - k|x - x'| = \bar{x}$, and since $\arg \max f_b(x) - k|x - x'| = \underline{x}$, we have: $\arg \max_{\hat{x} \in [0, 1]} \mathbb{E}[y|\hat{x}] - k|\hat{x} - x'| \in [\underline{x}, \bar{x}]$. All $x < \underline{x}$ are dominated by $\underline{x}$ and all $x > \bar{x}$ are dominated by $\bar{x}$.

Next, I claim $\arg \max_{\tilde{x} \in [0, 1]} W(q, \tilde{x}) - k|\tilde{x} - x'| \in [\underline{x}, \bar{x}]$. Suppose $x < \underline{x}$, meaning $f_b(x)$ is concave and $f'_b(x) > -k$ for $x < \underline{x}$. Then, the derivative with respect to x here is: $qk + (1 - q)f'_b(x) + k =$

$(1-q)(f'_b(x) + k) > 0$, meaning $\underline{x}$ dominates all options to its left. Similarly, for $x \in (\bar{x}, 1]$, $|f'(x)|$ is necessarily $< k$ and the derivative is $qf'(x) - (1-q)k - k < 0$.

Finally, suppose by contradiction that $[\underline{x}, \bar{x}] \not\prec \hat{x}$, where $\hat{x}$ is the maximizer associated with $W(q, x')$. If $\hat{x} < \underline{x}$, we know that $\mathbb{E}[y|\hat{x}] - k|\hat{x} - x'| < \mathbb{E}[y|\underline{x}] - k|\underline{x} - x'|$ and $\Phi(q, \hat{x}) - k|\hat{x} - x'| < \Phi(q, \underline{x}) - k|\underline{x} - x'|$. Any convex combination of $\mathbb{E}[y|\underline{x}] - k|\underline{x} - x'|$ and $\Phi(q, \underline{x}) - k|\underline{x} - x'|$ is strictly larger than any convex combination of $\mathbb{E}[y|\hat{x}] - k|\hat{x} - x'|$ and $\Phi(q, \hat{x}) - k|\hat{x} - x'|$, meaning $\bar{x}$ dominates $\hat{x}$ in the original program. $\square$

Lemma 5. Suppose $k < f'_g(x_{LT}) < 2k$. There exists $c > 0$ such that: $f_g(\bar{x}) - k(\bar{x} - x_{LT}) \geq y_{LT}$, $f_b(\underline{x}) - k(x_{LT} - \underline{x}) + c > y_{LT}$ and $W(1/2, x_{LT}) < y_{LT} + c$. Hence, the set $LT = \{(q, x') : W(q, x') < y_{LT} - c\}$ has nonempty interior.

Proof. We construct an upper bound for $W(1/2, x_{LT})$. First, I claim:

$$\begin{aligned} & \frac{1}{2}(f_g(\bar{x}) - k(\bar{x} - x_{LT})) + \frac{1}{2}(f_b(\underline{x}) - k(x_{LT} - \underline{x})) - k(\bar{x} - x_{LT}) < y_{LT} \\ & \frac{1}{2}(f_g(\bar{x}) - k(\bar{x} - x_{LT})) + \frac{1}{2}(f_b(\underline{x}) - k(x_{LT} - \underline{x})) - k(x_{LT} - \underline{x}) < y_{LT}. \end{aligned}$$

Note that because $f_b(x)$ and $f_g(x)$ are symmetric about $x_{LT} = \frac{1}{2}$, $f_g(\bar{x}) - k(x_{LT} - \bar{x}) = f_b(\bar{x}) - k(x_{LT} - \bar{x})$. Hence, the first equation reduces to $f_g(\bar{x}) - 2k(\bar{x} - x_{LT}) < y_{LT}$, which follows since $f'_g(x_{LT}) < 2k$.

We now construct our upper bound for $W(1/2, x_{LT})$. $W(1/2, x_{LT}) =$

$$\begin{aligned} & \max_{x \in [\underline{x}, \bar{x}]} \frac{2\sigma(1-\delta)\mathbb{E}[y|x] + \delta\Delta(x)\Phi(1/2, x)}{2\sigma(1-\delta) + \delta\Delta(x)} - k|x - x_{LT}| \\ &= \max_{x \in [\underline{x}, \bar{x}]} \frac{2\sigma(1-\delta)}{2\sigma(1-\delta) + \delta\Delta(x)} (\mathbb{E}[y|x] - k|x - x_{LT}|) + \frac{\delta\Delta(x)}{2\sigma(1-\delta) + \delta\Delta(x)} (\Phi(1/2, x) - k|x - x'|) \\ &\leq \max_{x \in [\underline{x}, \bar{x}]} \frac{2\sigma(1-\delta)}{2\sigma(1-\delta) + \delta\Delta(x)} \left(\max_{\check{x}} \mathbb{E}[y|\check{x}] - k|\check{x} - x_{LT}| \right) + \frac{\delta\Delta(x)}{2\sigma(1-\delta) + \delta\Delta(x)} (\Phi(1/2, x) - k|x - x'|) \\ &= \max_{x \in [\underline{x}, \bar{x}]} \frac{2\sigma(1-\delta)}{2\sigma(1-\delta) + \delta\Delta(x)} y_{LT} + \frac{\delta\Delta(x)}{2\sigma(1-\delta) + \delta\Delta(x)} (\Phi(1/2, x) - k|x - x_{LT}|) \equiv \max_{x \in [\underline{x}, \bar{x}]} U(x) \end{aligned}$$

$U(x)$ will be the objective associated with the upper bound. The third line holds since $\arg \max \mathbb{E}[y|\check{x}]$ when $q = \frac{1}{2}$ is simply $x_{LT} = \frac{1}{2}$. Moreover, note that Φ does not depend on x at $q = 1/2$: $1/2(f_g(\bar{x}) -$

$k(\bar{x} - x)) + 1/2(f_b(\underline{x}) - k(x - \underline{x})) = 1/2(f_g(\bar{x}) + f_b(\underline{x}) - k(\bar{x} - \underline{x}))$. Hence:

$$\begin{aligned}\Phi(1/2, x_{LT}) &= 1/2(f_g(\bar{x}) - k(\bar{x} - x_{LT})) + 1/2(f_b(\underline{x}) - k(x_{LT} - \underline{x})) > y_{LT} \\ &> 1/2(f_g(\bar{x}) - k(\bar{x} - x_{LT})) + 1/2(f_b(\underline{x}) - k(x_{LT} - \underline{x})) - k(\bar{x} - x_{LT}), \\ &1/2(f_g(\bar{x} - k(\bar{x} - x_{LT})) + 1/2(f_b(\underline{x}) - k(x_{LT} - \underline{x})) - k(\underline{x} - x_{LT})\end{aligned}$$

Let $\hat{x}_1$ and $\hat{x}_2$ be defined by the following:

$$\begin{aligned}y_{LT} &= 1/2(f_g(\bar{x} - k(\bar{x} - x_{LT})) + 1/2(f_b(\underline{x}) - k(x_{LT} - \underline{x})) - k(\hat{x}_1 - x_{LT}) & \hat{x}_1 \in (x_{LT}, \bar{x}) \\ y_{LT} &= 1/2(f_g(\bar{x} - k(\bar{x} - x_{LT})) + 1/2(f_b(\underline{x}) - k(x_{LT} - \underline{x})) - k(\hat{x}_2 - x_{LT}) & \hat{x}_2 \in (\underline{x}, x_{LT})\end{aligned}$$

which exist and are unique by continuity, strict monotonicity, and the Intermediate Value Theorem. Because we are taking convex combinations of y_{LT} and $\Phi - k|x - x_{LT}|$, where the weights depend on x as well, we will only want to place less weight on y_{LT} if $\Phi - k|x - x_{LT}| > y_{LT}$ (precisely on $[\hat{x}_2, x_{LT}]$ and $[x_{LT}, \hat{x}_1]$).

The derivative of $U(x)$ for $x > x_{LT}$ is:

$$\frac{2\sigma(1-\delta)\delta\Delta'(x)(\Phi - k(x - x_{LT}) - y_{LT}) - 2\sigma(1-\delta)\delta\Delta(x)k - \delta^2\Delta(x)^2k}{(2\sigma(1-\delta) + \delta\Delta(x))^2}$$

A nearly identical derivative for $x < x_{LT}$ is omitted. At $x = x_{LT}$, the right derivative is $\frac{\delta\Delta'(x)(\Phi - y_{LT})}{2\sigma(1-\delta)} > 0$. At $\hat{x}_1$, when $\Phi - k(\hat{x}_1 - x_{LT}) - y_{LT} = 0$, the derivative is $-\frac{\delta\Delta(x)k}{2\sigma(1-\delta) + \delta\Delta(x)} < 0$. The derivative is ≥ 0 when:

$$\begin{aligned}2\sigma(1-\delta)\delta\Delta'(x)(\Phi - k(x - x_{LT}) - y_{LT}) &\geq \delta\Delta(x)(2\sigma(1-\delta) + \delta\Delta(x))k \\ \Phi - k(x - x_{LT}) - y_{LT} &\geq \frac{\delta\Delta(x)(2\sigma(1-\delta) + \delta\Delta(x))k}{2\sigma(1-\delta)\delta\Delta'(x)} \\ \Phi - k(x - x_{LT}) &\geq y_{LT} + \frac{\Delta(x)(2\sigma(1-\delta) + \delta\Delta(x))}{2\sigma(1-\delta)\Delta'(x)}k \\ &= y_{LT} + k(x - x_{LT})\left(1 + \frac{\delta\Delta(x)}{2\sigma(1-\delta)}\right)\end{aligned}$$

The final line is strictly increasing in x with a zero at $x_{LT} < \hat{x}_1$. $\Phi - k(x - x_{LT})$ is strictly decreasing in x with a zero at $\hat{x}_1$. Hence there is a *unique* maximizer $\bar{x}^*$ to this problem in $(x_{LT}, \hat{x}_1)$. Define c

implicitly as $\Phi - k(\bar{x}^* - x_{LT}) = y_{LT} + c$.¹⁶ Then:

$$\begin{aligned} W(1/2, x_{LT}) &\leq \max_x U(x) = y_{LT} + \frac{\delta\Delta(\bar{x}^*)}{2\sigma(1-\delta) + \delta\Delta(\bar{x}^*)} \left(k(\bar{x}^* - x_{LT}) \left(1 + \frac{\delta\Delta(\bar{x}^*)}{2\sigma(1-\delta)} \right) \right) \\ &= y_{LT} + \frac{\delta\Delta(\bar{x}^*)}{2\sigma(1-\delta) + \delta\Delta(\bar{x}^*)} c < y_{LT} + c. \end{aligned}$$

Since $\bar{x}^* > x_{LT}$, we also have $\Phi(1/2, x_{LT}) - k(\bar{x}^* - x_{LT}) = y_{LT} + c \implies \Phi(1/2, x_{LT}) > y_{LT} + c$.

Since $W(q, x')$ is continuous, the inequality $W(1/2, x_{LT}) < y_{LT} + c$ also holds in a *neighborhood* of $(1/2, x_{LT})$. This means that $LT : \{(q, x') : W(q, x') \leq y_{LT} + c\}$ has nonempty interior. $\square$

Proof of Solution to Leader's Repeated Problem

We proceed with the proof of the leader's problem. Lemmata 6 and 7 lay out stationarity features of the no-overthrow set $NR(q, x')$. Lemma 8 illustrates what the leader would do if her choices were unconstrained. Theorem 1 generalizes Proposition 1 to the general case.

First, we define $\tilde{W}_{x_\ell(q, x')}$ recursively as follows:

$$\begin{aligned} \tilde{W}_{x_\ell(q, x')}(q, x') &= (1 - \delta)\mathbb{E}[y|x_\ell(q, x')] - k|x_\ell(q, x') - x'| + \frac{\delta\Delta(x)}{2\sigma}\tilde{\Phi}_{x_\ell(q, x')}(q, x_\ell(q, x')) \\ &\quad + \delta\left(1 - \frac{\Delta(x)}{2\sigma}\right)\tilde{W}_{x_\ell(q, x')}(q, x_\ell(q, x')) \\ \tilde{\Phi}_{x_\ell(q, x')}(q, x) &= q\tilde{W}(1, x_\ell(1, x)) + (1 - q)\tilde{W}(1, x_\ell(0, x)) \end{aligned}$$

This represents the *people's* value from accepting a leader's policy plan $x_\ell(q, x')$. If the leader's strategy is only to implement x_{LT} for all time, the value of this objective is $y_{LT} - k|x_{LT} - x'|$. The following lemmata provide a closed form for this expression and show that, in equilibrium, this strategy has the property that $x_\ell(q, x) = x_\ell(q, x_\ell(q, x))$.

Lemma 6. *Let $x_\ell(q, x')$ denote a policy for the leader such that $x_\ell(q, x_\ell(q, x')) = x_\ell(q, x')$. Then, the people overthrow if and only if $\tilde{W}(x_\ell, q, x') \geq W(q, x') - c$, where*

$$\tilde{W}(x_\ell, q, x') = \frac{2\sigma(1-\delta)\mathbb{E}[y|x_\ell(q, x')] + \delta\Delta(x_\ell(q, x'))\tilde{\Phi}(q, x_\ell(q, x'))}{2\sigma(1-\delta) + \delta\Delta(x)} - k|x_\ell(q, x') - x'|$$

¹⁶Defining c using the derivative to the left of x_{LT} gives the same c , since Φ , $\Delta(x)$, and $k|x - x_{LT}|$ are symmetric about x_{LT} .

where $\tilde{\Phi}(q, x) = q(f_g(x_\ell(1, x) - k|x_\ell(1, x) - x'|) + (1 - q)(f_b(x_\ell(0, x) - k|x_\ell(0, x) - x|))$.

Proof. Let $\tilde{W}(x_\ell, q, x')$ be the value for the people of accepting for all time a policy $x_\ell(q, x')$ satisfying the hypothesis. Let

$$\begin{aligned}\tilde{\Phi}(q, x) &= q\tilde{W}(x_\ell, 1, x_\ell(1, x_\ell(1, x')) + (1 - q)\tilde{W}(x_\ell, 1, x_\ell(0, x_\ell(0, x')) \\ \tilde{W}(x_\ell, 1, x') &= \left(\sum_{t=0}^{\infty} (1 - \delta)f_g(x_\ell(1, x')) \right) - k|x_\ell(1, x') - x'| = f_g(x_\ell(1, x') - k|x_\ell(1, x') - x'| \\ \tilde{W}(x_\ell, 0, x') &= \left(\sum_{t=0}^{\infty} (1 - \delta)f_b(x_\ell(0, x')) \right) - k|x_\ell(0, x') - x'| = f_b(x_\ell(0, x') - k|x_\ell(0, x') - x'|.\end{aligned}$$

Then:

$$\begin{aligned}\tilde{W}(x_\ell, q, x_\ell(q, x')) &= (1 - \delta)\mathbb{E}[y|x_\ell(q, x')] + \delta \frac{\Delta(x_\ell(q, x'))}{2\sigma} \Phi(q, x_\ell(q, x')) \\ &\quad + \delta \left(1 - \frac{\Delta(x)}{2\sigma}\right) \tilde{W}(q, x_\ell(q, x')) \\ &= \frac{2\sigma(1 - \delta)\mathbb{E}[y|x_\ell(q, x')] + \delta\Delta(x_\ell(q, x'))\Phi(q, x_\ell(q, x'))}{2\sigma(1 - \delta) + \delta\Delta(x)}\end{aligned}$$

Replicating arguments from earlier lemmata gives the result:

$$\tilde{W}(x_\ell, q, x') = \frac{2\sigma(1 - \delta)\mathbb{E}[y|x_\ell(q, x')] + \delta\Delta(x_\ell(q, x'))\Phi(q, x_\ell(q, x'))}{2\sigma(1 - \delta) + \delta\Delta(x)} - k|x_\ell(q, x') - x'|$$

□

Let $\text{NR}(q, x')$ be the set of policies that weakly prevents credible overthrow:

$$\text{NR}(q, x') = \{x \in [0, 1] : \tilde{W}_x(q, x') \geq W(q, x') - c\}.$$

As before, the people overthrow the leader if and only if $x_\ell(q, x') \notin \text{NR}(q, x')$.

Lemma 7. $x_\ell(q, x') = x_\ell(q, x_\ell(q, x'))$, meaning the previous lemma characterizes the equilibrium overthrow decision.

Proof. If $\text{NR}(q, x')$ does not bind, the leader implements her preferred policy by the previous lemma, which satisfies the property. So, suppose $\text{NR}(q, x')$ binds and that by contradiction there exists x' such that $x_1 \equiv x_\ell(q, x') \neq x_\ell(x_\ell(q, x')) \equiv x_2$. Let $h(x) = (1 - \delta)\mathbb{E}[y|x] + \frac{\delta\Delta(x)}{2\sigma}\tilde{\Phi}(q, x)$. Then,

we must have

$$\begin{aligned}
W(q, x') - c &= h(x_1) - k|x_1 - x'| \\
&\quad + \delta\left(1 - \frac{\Delta(x_1)}{2\sigma}\right)\left(h(x_2) + \delta\left(1 - \frac{\Delta(x_2)}{2\sigma}\right)\tilde{W}_{x_\ell}(x_2) - k|x_2 - x_1|\right) \\
W(q, x_1) - c &= h(x_2) + \delta\left(1 - \frac{\Delta(x_2)}{2\sigma}\right)\tilde{W}_{x_\ell}(x_2) - k|x_2 - x_1|
\end{aligned}$$

Let $x^* = x_p(q, x')$. Since $\text{NR}(q, x')$ is binding, it must be that:

$$\begin{aligned}
&\frac{2\sigma(1-\delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q, x^*)}{2\sigma(1-\delta) + \delta\Delta(x^*)} - k|x^* - x'| - c > \\
&\quad h(x_2) + \delta\left(1 - \frac{\Delta(x_2)}{2\sigma}\right)\tilde{W}_{x_\ell}(x_2) - k|x_2 - x'|
\end{aligned}$$

where the RHS is equal to $W(q, x')$. Since either $x^* < x_1 < x_2$ or $x^* > x_1 > x_2$ by a previous lemma (policy is attracted in a single direction; implementing x^* and then x_2 is dominated by a policy between x^* and x_2), as $x' \rightarrow x_1$, both sides increase/decrease by the same amount or the right side increases by more, meaning:

$$\frac{2\sigma(1-\delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q, x^*)}{2\sigma(1-\delta) + \delta\Delta(x^*)} - k|x^* - x_1| > h(x_2) + \delta\left(1 - \frac{\Delta(x_2)}{2\sigma}\right)\tilde{W}_{x_\ell}(x_2) - k|x_2 - x_1|$$

But then:

$$\begin{aligned}
W(q, x_1) &= \max_{x \in [0,1]} \frac{2\sigma(1-\delta)\mathbb{E}[y|x] + \delta\Delta(x)\Phi(q, x)}{2\sigma(1-\delta) + \delta\Delta(x)} - k|x - x_1| \\
&\geq \frac{2\sigma(1-\delta)\mathbb{E}[y|x^*] + \delta\Delta(x^*)\Phi(q, x^*)}{2\sigma(1-\delta) + \delta\Delta(x^*)} - k|x^* - x_1|
\end{aligned}$$

a contradiction.

□

To formally characterize the leader's solution, we introduce the following notation:

$$\begin{aligned}
\underline{x}_{LT}(q, x') &= \max\{x \in \text{NR}(q, x') : x \leq x_{LT}\} & \bar{x}_{LT}(q, x') &= \min\{x \in \text{NR}(q, x') : x \geq x_{LT}\} \\
\underline{x}(q, x') &= \min\{x \in \text{NR}(q, x')\} & \bar{x}(q, x') &= \max\{x \in \text{NR}(q, x')\}
\end{aligned}$$

$\underline{x}_{LT}$ and $\bar{x}_{LT}$ correspond to the *closest* policies to the learning trap preventing overthrow. $\bar{x}$ and $\underline{x}$ are the lowest and highest policies preventing overthrow altogether. As before, we also have:

$$\underline{u}(x') \equiv u_\ell(x_\ell(0, x')) < u_\ell(y_{LT}) < u_\ell(x_\ell(1, x')) \equiv \bar{u}(x').$$

The Bellman describing the leader's optimal solution $x_\ell(q, x')$ is then:

$$V(q, x') = \max_{x_\ell(q, x') \in \text{NR}(q, x')} (1 - \delta^\ell) u_\ell(x) + \frac{\delta^\ell \Delta(x)}{2\sigma} \Psi(q, x) + \delta \left(1 - \frac{\delta^\ell \Delta(x)}{2\sigma}\right) V(q, x)$$

where $\Psi(q, x') = q\bar{u}(x') + (1 - q)\underline{u}(x')$.

Lemma 8. *Suppose that the leader can implement any $x \in [x_\ell(0, x'), x_\ell(1, x')]$. Then, there exist thresholds $\underline{q} < \bar{q}$ such that: for $q \leq \underline{q}$, the derivative of the objective in the maximization problem is positive for $x < x_{LT}$ and negative for $x > x_{LT}$; for $q \in [\underline{q}, \bar{q}]$, the derivative of the objective is positive for $x < x_{LT}$ and positive for $x > x_{LT}$; and for $q \geq \bar{q}$, the derivative of the objective is negative for $x < x_{LT}$ and positive for $x > x_{LT}$.*

Proof. Let $X_{unc} = [x_\ell(0, x'), x_\ell(1, x')]$. The leader's Bellman in the unconstrained case can be written as:

$$\tilde{V}(q) = \max_{x \in X_{unc}} (1 - \delta) u_\ell(x) + \delta \frac{\Delta(x)}{2\sigma} (q\bar{u}(x') + (1 - q)\underline{u}(x')) + \delta \left(1 - \frac{\Delta(x)}{2\sigma}\right) \tilde{V}(q)$$

Let $\Psi(q) = q\bar{u}(x') + (1 - q)\underline{u}(x')$. Replicating an earlier argument implies

$$\tilde{V}(q) = \max_{x \in X_{unc}} \frac{2\sigma(1 - \delta)u_\ell(x) + \delta\Delta(x)\Psi(q)}{2\sigma(1 - \delta) + \delta\Delta(x)}.$$

The numerator of the derivative with respect to x for $x \neq x_{LT}$ is:

$$= 4\sigma^2(1 - \delta)^2 u'_\ell(x) + 2\sigma\delta(1 - \delta)(\Delta(x)u'_\ell(x) + \Delta'(x)\Psi(q) - \Delta'(x)u_\ell(x)).$$

Observe that $\Delta(x)$ is the absolute value of a linear function with zero at $x_{LT} = \frac{1}{2}$.

$$\Delta(x) = |-(x - 1)^2 - x^2| = |-1 + 2x| \equiv \xi|x - x_{LT}|.$$

This means that for $x > x_{LT}$, the derivative of the objective is negative if and only if

$$2\sigma(1-\delta)u'_\ell(x) \leq \delta(\Delta'(x)u_\ell(x) - \Delta(x)u'_\ell(x) - \Delta'(x)\Psi(q))$$

$$2\sigma(1-\delta)u'_\ell(x) \leq \delta\xi(u_\ell(x) - u'_\ell(x)(x - x_{LT}) - \Psi(q))$$

$u_\ell(x) - u'_\ell(x)(x - x_{LT})$ is equal to x_{LT} at $x = x_{LT}$ and is weakly increasing for $x > x_{LT}$.¹⁷ By weak concavity, the LHS below is decreasing and the RHS increasing:

$$2\sigma(1-\delta)u'_\ell(x) \leq \delta\xi(u_\ell(x) - u'_\ell(x)(x - x_{LT}) - \Psi(q)) \quad (12)$$

If $\sigma(1-\delta)$ is small, since $\underline{u}(x') < x_{LT} \leq u_\ell(x) - u'_\ell(x)(x - x_{LT}) < \bar{u}(x')$, there is a threshold $\underline{q}$ such that the inequality satisfies strictly if and only if $q < \underline{q}$. Moreover, for any $\sigma(1-\delta)$, as $\underline{u}(x')$ grows more negative, $-\Psi(q)$ grows large, which can also allow the inequality to satisfy. A similar argument shows that for $x < x_{LT}$, the derivative is positive when:

$$2\sigma(1-\delta)u'_\ell(x) \geq \delta\xi(\Psi(q) - (u_\ell(x) - u'_\ell(x)(x - x_{LT}))) \quad (13)$$

When (12) holds with equality, its RHS is strictly positive, meaning the RHS of (13) is strictly negative. Letting $\bar{q}$ be the point where (13) holds with equality, we thus have that $\underline{q} < \bar{q}$. Hence: for $x > x_{LT}$, the derivative of the objective is strictly negative when $q < \underline{q}$ and strictly positive when $q > \bar{q}$, with equality at $q = \underline{q}$; for $x < x_{LT}$, the derivative of the objective is strictly positive when $q < \bar{q}$ and strictly negative when $q > \bar{q}$, with equality at $q = \bar{q}$. This completes the lemma. $\square$

Theorem 1. Suppose $\sigma(1-\delta^\ell)$ is small or $\underline{u}(x')$ is sufficiently low. Then, there exist thresholds $\underline{q} < \bar{q} \in (0, 1)$ such that the following hold:

1. If $q \leq \underline{q}$, the leader plays the closest policy to the learning trap (which may be greater or less than x_{LT}).
2. If $q \in [\underline{q}, \bar{q}]$, the leader plays either the lowest policy below the learning trap or her highest feasible policy.

¹⁷To see this, note that the derivative of this expression with respect to x is $u'_\ell(x) - u''_\ell(x)(x - x_{LT}) - u'_\ell(x) = -u''_\ell(x)(x - x_{LT}) \geq 0$.

3. If $q > \bar{q}$, the leader either plays her highest feasible policy or lowest feasible policy.

The hypotheses of the theorem are described in terms of the above notation as follows.

- If $q \leq \underline{q}$, then $x_\ell(q, x') = \underline{x}_{LT}(q, x')$ or $\bar{x}_{LT}(q, x')$.
- If $q \in [\underline{q}, \bar{q}]$, then $x_\ell(q, x') = \underline{x}_{LT}(q, x')$ or $\bar{x}(q, x')$.
- If $q > \bar{q}$, then $x_\ell(q, x') = \bar{x}(q, x')$ or $\underline{x}(q, x')$.

Proof. We use the lemmata above to prove the original theorem; the solution can be written:

$$V(q, x') = \max_{x \in \text{NR}(q, x')} \frac{2\sigma(1-\delta)u_\ell(x) + \delta\Delta(x)\Psi(q)}{2\sigma(1-\delta) + \delta\Delta(x)}$$

where, replicating the previous lemma's argument, we have $\Psi(q) = q\bar{u}(x') - (1-q)\underline{u}(x')$. If $q \leq \underline{q}$, the derivative for $x > x_{LT}$ is negative and the derivative for $x < x_{LT}$ is positive, meaning the leader is attracted to the learning trap. Hence, $x_\ell(q, x') = \underline{x}_{LT}(q, x')$ or $\bar{x}_{LT}(q, x')$. If $q \in [\underline{q}, \bar{q}]$, the solution is either $\underline{x}_{LT}(q, x')$ or some $x \geq x_{LT}$. If $q \geq \bar{q}$, the derivative for $x > x_{LT}$ is positive and the derivative for $x < x_{LT}$ is negative. Hence the optimal solution is either $\bar{x}(q, x')$ or $\underline{x}(q, x')$. $\square$

Additional Results

Model with Non-Uniform Distribution of ϵ_t

Proposition 4. *Suppose the graph of $\text{NR}(q)$ is convex. There exists a threshold $\underline{q} \in (0, 1]$ such that:*

1. *if $q \leq \underline{q}$, $x_\ell(q) = \bar{x}(q)$. The leader plays the most extreme policy preventing overthrow.*
2. *If $q \geq \underline{q}$, the leader implements $x_\ell(q) \in [\min\{x_{LT}, \underline{x}(q)\}, \bar{x}(q)]$. Moreover, for δ high, $x_\ell(q) < \bar{x}(q)$; i.e., she moderates her actions to pursue the learning trap.*

Proof. Suppose q is such that $\bar{x}(q) \leq 1$. Because the graph of $\text{NR}(q) = [q - \sqrt{c}, \min\{q + \sqrt{c}, 1\}]$ is convex, by Theorem 9.8 in Stokey et al. (1989), $V(q)$ is concave in q .

Let $\underline{q} \in (0, 1)$ solve $\underline{q} + \sqrt{c} = x_{LT}$. Suppose $q \leq \underline{q}$. Note that, on this range, an increase from x to x' has two effects. First, it increases flow utility $u_\ell(q)$. Second, because $q \leq \underline{q}$, we must have $x_{LT} \geq x' > x$, so $|x - x_{LT}| > |x' - x_{LT}|$, meaning $\tau(q'|q, x')$ is a mean-preserving contraction of $\tau(q'|q, x)$. This means that, because $V(q)$ is concave in q , $\int V(q')\tau(q'|q, x') > \int V(q')\tau(q'|q, x)$. Hence, the leader will pursue $\bar{x}(q)$, which is both the highest possible policy and the closest policy to the learning trap.

Next, suppose $q > \underline{q}$. By the previous argument, all feasible $x < x_{LT}$ are dominated by x_{LT} . Now, above x_{LT} , an increase from x to x' has two effects. First, it increases flow utility: $u_\ell(x') > u_\ell(x)$. Second, because V is concave, it *decreases* the value of continuation utility, i.e. $\int V(q')\tau(q'|q, x') < \int V(q')\tau(q'|q, x)$. For δ large, this second informational effect will tend to outweigh the former flow effect, suggesting that $\min\{\underline{x}(q), x_{LT}\} \leq x_\ell(q) < \bar{x}(q)$. $\square$

Three States of the World

This section establishes an analogue of Lemma 8 for three states of the world, since deriving properties of $\text{NR}(q)$ are challenging. The three states of the world are g, m, b . The data-generating

processes in state i is given by f_i :

$$\begin{aligned} f_g(x) \quad y_t &= -(x_t - 1)^2 + \epsilon_t \\ f_m(x) \quad y_t &= -(x_t - \tilde{x}_m)^2 + \epsilon_t \\ f_b(x) \quad y_t &= -x_t^2 + \epsilon_t. \end{aligned}$$

for $\tilde{x}_m \in (0, 1)$ and $\epsilon_t \sim \mathcal{U}[-\sigma, \sigma]$. We assume $f_i(x) - f_j(x)$ each single-cross at some x_{LT}^{ij} for all $i \neq j$. The state variables are q_g (belief that g holds) and q_m (belief that m holds). We fix $q_g, q_m > 0$ and $q_g + q_m < 1$ for the remainder of the analysis. Beginning at a vector of states (q_g, q_m) , we can write the following expected values of the problem, as well as the probabilities with which they occur (which are all functions of x).

- V^{gmb} : initial value of the problem at (q_g, q_m) .
- V^g : value of problem if g is true; $(q_g, q_m) \rightarrow (1, 0)$; occurs with prob. p_g . Solution is denoted x_ℓ^g .
- V^m : value of problem if m is true; $(q_g, q_m) \rightarrow (0, 1)$; occurs with prob. p_m . Solution is denoted x_ℓ^m .
- V^b value of problem if b is true; $(q_g, q_m) \rightarrow (0, 0)$; occurs with prob. p_b . Solution is denoted x_ℓ^b .
- V^{gb} value of problem if only m is not true; $(q_g, q_m) \rightarrow (\frac{q_g}{1-q_m}, 0)$; occurs with prob. p_{gb} .
- V^{gm} value of problem if only b is not true; $(q_g, q_m) \rightarrow (\frac{q_g}{q_g+q_m}, \frac{q_m}{q_g+q_m})$; occurs with prob. p_{gm} .
- V^{mb} value of problem if only g is not true; $(q_g, q_m) \rightarrow (0, \frac{q_m}{1-q_g})$; occurs with prob. p_{mb} .

Each of these probabilities can be computed explicitly in terms of the beliefs and x ; for example, $p_g(x)$ is given by $q_g \cdot \frac{|f_g(x) - f_b(x)|}{2\sigma} \cdot \frac{|f_g(x) - f_m(x)|}{2\sigma}$.

Note that $x_\ell^b < x_{LT}^{gb} < x_\ell^g$ and $x_\ell^b < x_{LT}^{mb} < x_\ell^m$. Fixing $1 - q_m - q_b$, suppose the difference between $V^b = u_\ell(x_\ell^b)$ and $V^m = u_\ell(x_\ell^m)$ is sufficiently large. Applying Lemma 8 shows:

- $V^{gb} = u_\ell(x_{LT}^{gb})$: suppressing the risk of information revelation that could reveal b as true, the leader will choose a “pairwise” learning trap x_{LT}^{gb}

- $V^{mb} = u_\ell(x_{LT}^{mb})$: suppressing the risk of information revelation that could reveal b as true, the leader will choose a “pairwise” learning trap x_{LT}^{mb}

This gives us the following result:

Lemma 9. Suppose that $q_g, q_m > 0$ and $q_g + q_m < 1$; the leader’s policies are constrained if g , m , or b are revealed as true but can otherwise play any policy in $[x_\ell^b, x_\ell^g]$; and V^b is sufficiently low. Then, the leader either implements a learning trap policy x_{LT}^{gb} or x_{LT}^{mb} .

Proof. Replicating earlier arguments, the leader’s problem can be written as a convex combination of flow and continuation values:

$$V^{gmb} = \max_x \frac{(1-\delta)u_\ell(x) + \delta(p_g V^g + p_m V^m + p_b V^b + p_{gb} V^{gb} + p_{gm} V^{gm} + p_{mb} V^{mb})}{(1-\delta) + \delta(p_g + p_m + p_b + p_{gb} + p_{gm} + p_{mb})}$$

$$V^{gmb} = \max_x \frac{(1-\delta)u_\ell(x) + \delta(p_g V^g + p_m V^m + p_b V^b + p_{gb} u_\ell(x_{LT}^{gb}) + p_{gm} V^{gm} + p_{mb} u_\ell(x_{LT}^{mb}))}{(1-\delta) + \delta(p_g + p_m + p_b + p_{gb} + p_{gm} + p_{mb})}$$

Since we assume V^b is sufficiently small, the leader will do anything in her power to suppress revealing that b is true, i.e. will try its best to set $p_b = 0$. p_b is expressed as:

$$p_b = (1 - q_g - q_m) \frac{|f_g(x) - f_b(x)|}{2\sigma} \cdot \frac{|f_m(x) - f_b(x)|}{2\sigma}$$

With probability $1 - q_g - q_m$, b is true. Then, with probability $\frac{|f_g(x) - f_b(x)|}{2\sigma}$, state b is separable from state g and with probability $\frac{|f_m(x) - f_b(x)|}{2\sigma}$ it is separable from state m . Because the left and right derivatives of p_b are always nonzero, the only way to set p_b equal to zero is either to set $f_g(x) = f_b(x)$ by playing x_{LT}^{gb} , or to set $f_m(x) = f_b(x)$ by implementing x_{LT}^{mb} . In the former case, either m is revealed as the state or the people are uncertain about the state being b or g ; the value for the leader is:

$$\frac{(1-\delta)u_\ell(x_{LT}^{gb}) + \delta(p_m V^m + p_{gb} u_\ell(x_{LT}^{gb}))}{(1-\delta) + \delta(p_m + p_{gb})}.$$

In the latter case, either g is revealed as the state or the people are uncertain between b and m ; the

value for the leader is:

$$\frac{(1 - \delta)u_\ell(x_{LT}^{mb}) + \delta(p_g V^g + p_{mb} u_\ell(x_{LT}^{mb}))}{(1 - \delta) + \delta(p_g + p_{mb})}.$$

If q_g is sufficiently large, the leader will choose to implement x_{LT}^{mb} , which favors her in expectation; otherwise she will choose x_{LT}^{gb} .

The option that is chosen will depend on the parameters of the model (precise beliefs and values of x_{LT}^{gb} and V^m). $\square$

Finally, the terminal history of the model always reduces to one or two states of the world. Suppose the leader implements x_{LT}^{gb} , allowing information revelation only about whether $\{g, b\}$ or $\{m\}$ is true. In the former case, we arrive at the baseline two-state model. On the equilibrium path, uncertainty is thus resolved only upon the revelation of states to which the leader is not averse.

Model with Ambiguity

Proposition 5. *Let $\mathcal{Q} \subseteq [0, 1]$ be a compact set of constituent beliefs such that $\mathcal{Q} \cap (0, 1) \neq \emptyset$. Suppose the leader operates as a maxmin expected utility maximizer and δ is large. Let $x_\ell(\mathcal{Q})$ be the leader's optimal policy conditional on believing the people's belief q lies in $\mathcal{Q}$. Then:*

- *if $x_{LT} \in \text{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q}) = x_{LT}$;*
- *if there exists x such that $x \in \text{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q})$ is the highest policy that prevents overthrow;*
- *if there does not exist $x \in [0, 1]$ such that $x \in \text{NR}(q)$ for all $q \in \mathcal{Q}$, $x_\ell(\mathcal{Q}) = 1$.*

Proof. Define $\text{NR}(\mathcal{Q}) \equiv \bigcap_{q \in \mathcal{Q}} \text{NR}(q)$. Because $\text{NR}(q)$ is closed for each q , $\text{NR}(\mathcal{Q})$ is also closed, but may be empty. As before, we normalize all flow utilities to $(1 - \delta)u_\ell(x)$.

Suppose the leader implements $x \neq x_{LT}$. Since the anti-leader state of the world realizes with positive probability, the leader evaluates her continuation value as $\underline{u}$, meaning her value from pursuing x is $(1 - \delta)u_\ell(x) + \delta\underline{u}$. This is maximized at $u_\ell(1) = 1$. If the leader implements $x = x_{LT}$, no information is revealed. Thus, if $x_{LT} \in \text{NR}(\mathcal{Q})$ today, it is also in no-overthrow constraint

tomorrow, meaning the utility from pursuing x_{LT} is x_{LT} . Since δ is large, $x_{LT} \geq (1 - \delta) + \delta \underline{u}$. Hence, if $x_{LT} \in \text{NR}(\mathcal{Q})$, the leader pursues x_{LT} .

If $x_{LT} \notin \text{NR}(\mathcal{Q})$ but there exists x such that $x \in \text{NR}(q)$ for all $q \in \mathcal{Q}$, the leader's value is given by $\max_{x \in \text{NR}(q)} (1 - \delta)u_\ell(x) + \delta \underline{u}$. This is maximized at $x = \max\{\text{NR}(\mathcal{Q})\}$.

Finally, suppose $\text{NR}(\mathcal{Q})$ is empty. This means any x the leader pursues could risk overthrow, meaning her policy choices do not matter, and pursuing $x = 1$ is an equilibrium. $\square$

External Information Revelation

Proposition 6. *Suppose η increases. Then, $\underline{q}$ decreases and $\bar{q}$ increases. The leader plays the learning trap for fewer values of q .*

Proof. The objective of an unconstrained problem here can be written as:

$$\max_{x \in [0,1]} \frac{2\sigma(1 - \delta)u_\ell(x) + \delta((1 - \eta)\Delta(x) + 2\sigma\eta)\Psi(q)}{2\sigma(1 - \delta) + \delta((1 - \eta)\Delta(x) + 2\sigma\eta)}$$

The numerator of the derivative can be expressed as:

$$\begin{aligned} &= (2\sigma(1 - \delta) + \delta(1 - \eta)\Delta(x) + 2\sigma\delta\eta)(2\sigma(1 - \delta)u'_\ell(x) + \delta(1 - \eta)\Delta'(x)\Psi(q)) \\ &\quad - \delta(1 - \eta)\Delta'(x)(2\sigma(1 - \delta)u_\ell(x) + \delta((1 - \eta)\Delta(x) + 2\sigma\eta)\Psi(q)) \\ &= 4\sigma^2(1 - \delta)^2u'_\ell(x) + 2\sigma\delta(1 - \delta)(1 - \eta)(\Delta(x)u'_\ell(x) + \Delta'(x)\Psi(q) - \Delta'(x)u_\ell(x)) \\ &\quad + 2\sigma\delta\eta(2\sigma(1 - \delta)u'_\ell(x) + \delta(1 - \eta)\Delta'(x)\Psi(q) - (1 - \eta)\Delta'(x)\Psi(q)) \\ &= 4\sigma^2(1 - \delta)^2u'_\ell(x) + 2\sigma\delta(1 - \delta)(1 - \eta)(\Delta(x)u'_\ell(x) + \Delta'(x)\Psi(q) - \Delta'(x)u_\ell(x)) \\ &\quad + 4\sigma\delta(1 - \delta)\eta u'_\ell(x) \end{aligned}$$

For $x > x_{LT}$, the derivative is negative, replicating a previous lemma, if:

$$2\sigma(1 - \delta)u'_\ell(x) + 2\sigma\delta\eta u'_\ell(x) \leq \delta(1 - \eta)\xi(u_\ell(x) - u'_\ell(x)(x - x_{LT}) - \Psi(q))$$

where $\underline{q}$ is the unique point that solves this equation with equality. For each x , the LHS is increasing in η and the RHS is decreasing in η . This means that as η increases from 0, $\underline{q}$ shifts to the left.

The equation defining $\bar{q}$ can be written as:

$$2\sigma(1-\delta)u'_\ell(x) + 2\sigma\delta\eta u'_\ell(x) = \delta(1-\eta)\xi\left(\Psi(\bar{q}) - \left(u_\ell(x) - u'_\ell(x)(x - x_{LT})\right)\right)$$

An increase in η likewise causes the LHS to increase and RHS to decrease, causing $\bar{q}$ to increase. $\square$

Additional Case Study Material

Extreme Equilibrium of Model: Stolypin Reforms

The main text argues that when there was uncertainty about whether liberal reform was aligned with constituents' interests, the Russian Tsar pursued a learning trap; indeed, few radical modifications to the post-1861 arrangement occurred over the ensuing decades. This section argues that when there was more *certainty* that liberal reform would benefit Russians, the Tsar's government experimented with private property under the programs of Prime Minister Pyotr Stolypin beginning in 1906.

These reforms were motivated by two forces that represent a large shock to beliefs in favor of liberal reform. The first was the culmination of investigations into the affairs of the peasantry led by Sergei Witte, the Russian Minister of Finance, whose findings — along with independent recommendations of the Ministry of Internal Affairs — favored the privatization of property and “individualized farming” (Yaney, 1964, 278). The second was the Russian Revolution of 1905 — peasant uprisings that led to the establishment of (limited) consitutional monarchy (Polunov et al., 2015, 223) and the suspension of households' redemption payments, making the reformation of social order — changing the relationship of the peasantry to the land — necessary to complement these new political institutions; indeed, the reforms existed “within the context of other legislative and organizational measures that Nicholas II's government introduced” (Pallot, 1999, 3). The confluence of these forces meant that “a sufficient number of officials in Nicholas II's government shared this conviction [that individualized farming should be introduced into the countryside] for it to find its way into official state policy” (Pallot, 1999, 4).

Stolypin's 1906 Reform gave households the choice between holding property in private or communal tenure. Households could leave the commune while maintaining allotments or an equivalent parcel of land, overseen by a state-mediated bargaining process. The dissolution of the commune's power was accompanied by a reduction in the enforcement of mobility restrictions. Stolypin indeed believed that peasants who found themselves landless would find industrial work in the cities. From 1906 until the outbreak of the First World War, over one third of peasant households withdrew from the commune (Polunov et al., 2015, 231-32).

Stolypin's reforms were seen as experiments. The means of achieving land enclosure and

alignment of rural social order with politically liberal institutions evolved as officials learned about peasants' willingness to take up the reform and grappled with administration (Yaney, 1964, 275-276, 283). The culmination of the process would, ideally, result in economic growth for Russia, and "[o]nce enlightened through education and example to the possibilities of organizing their farms differently, it was thought that the majority of Russia's peasants would greet the reform enthusiastically" (Pallot, 1999, 5), creating "an independent, prosperous husbandman, a stable citizen of the land" (1). Thus, through the lens of the model, we can view the Stolypin reforms as departures from the learning trap triggered by an exogenous change in constituents' beliefs that favored the applicability of liberal reform to Russia. While the eventual course of the reform was cut short by the First World War, many argued that these reforms put Russia on the path towards economic development in imitation of the Western European model. Dower and Markevich (2019) argues that, indeed, the reform had a positive effect on the productivity of land-use in its short lifespan.

Alternative Explanations for Learning Trap

The literature discusses two main *non-informational* arguments motivating the "middle ground" achieved by the Tsar's 1861 reforms. First, this mixture may have balanced the desires of elites with those of the state. Second, Russia's lack of state capacity may have prevented more radical reform. In this section, I show that while these factors undoubtedly affected the emancipation reforms, the threat of informational revelation still shaped the Tsar's policy choices, and the explanation of a learning trap can be seen as complementary to these other stories.

Elite Power The landed nobility's commitment to serfdom constrained previous attempts to abolish serfdom. Could the post-1861 reforms have simply been an outcome of bargaining between a liberal State and pro-serfdom nobles?

On the one hand, the commune may have been logistically preferable for nobles to deal with civil and property disputes (Khristoforov and Gilley, 2016, 12). Labor allocation and rental payments may have been easier for landlords to organize when serfs were held collectively responsible (Dennison, 2014, 257). Landlords often disapproved of arrangements forcing them to give land to peasants, and extensively bargained over the size of government compensation during the

years of the reform (Moon, 2014, chaps. 7-8). Nevertheless, the constraint binding the government *to the commune* was far weaker in the 1850s than prior.

However, nobles, especially in the 1850s, were ideologically diverse, and many nobles' preferences aligned with the liberal aspirations of the Tsar. Most educated Russians believed that "the sanctity of private property was the basis of the political and social order" (Field, 1976, 58). Some liberals supported abolishing serfdom and endowing peasants with land. Even some conservative nobles wanted peasants to rent and work noble land in a competitive market. Many critics of the post-1861 system — at times more critical of serfdom than the government's own reformers — contrasted Russia's system of land tenure with the more desirable British case, "where large-scale land tenure and local self-government controlled by the aristocracy allegedly guaranteed economic prosperity and political stability" (Khristoforov, 2009, 58, 63-65). Khristoforov (2022) goes as far as to argue that some form of "private property individualism" was in the interest of both landlords and the government.¹⁸

Finally, a sufficient bloc of the nobility may not have wanted to challenge the Tsar; they "did not want political change... They wanted the tsar's favor... At each stage in the development of the government's program, the nobility swallowed its objections or stated them obliquely" (Field, 1976, 362). Thus not only may liberal reform may not have been "at a cost" to the nobility, even if it was, it is not clear that they had much power to usurp the Tsar's decisionmaking.

State Capacity State capacity undoubtedly constrained the Tsar's reform ambitions. Emancipating and endowing peasants with property, compensating nobles, setting up a cadastre, and securing property rights was expensive. Depressed stock prices and banking crises in the 1850s drained coffers (Dennison, 2020, 197), meaning the government had to look to the gentry for help. Landlords then played a large role in commune politics after emancipation (Dennison, 2020); outsourcing dispute resolution and tax collection to the commune was logistically easier than protecting individual household rights (Khristoforov (2022, S163), Dennison (2011)).

Puzzles still arise from ascribing the intermediacy of the reform solely to weak administration. First, before and after the 1850s, the government could have relaxed the commune's power on State lands, where elites had a much weaker hold. Instead, the institutional experience of the State

¹⁸One reason was that a liberal system could oppose socialism.

peasantry faced few changes before and after 1861, and outcomes of ex-Serf and State peasants converged by 1900 (Nafziger, 2012). Second, the State made few attempts to demarcate property. Officials wanted to tie peasants to the land, knew private property was easier to abandon through sale or mortgage, and in fact refused to separate the tax liabilities of some peasants who paid off their redemption obligations early (Khristoforov, 2007, 30). Dennison (2006) shows how the post-1861 commune sustained a pre-serfdom informal economy of social networks and patronage, even though the state could have slowly introduced “universal enforcement of contracts and property rights” (89).

The state’s nebulous attitude towards demarcation emanated in part from an informational desire to limit market activity. The results of a project attempting to construct a land cadastre prior to 1861 under Kiselev were never clearly revealed or publicly accessible, likely because the results may not have favored rationalization of property (Khristoforov and Gilley, 2016, 9, 11). Fourth, while the Tsar’s committees “recognized [the commune] as a temporary fact” (Khristoforov, 2007, 30), Alexander II never took further steps towards dissolution of the commune, and his son infamously strengthened its enforcement powers¹⁹ (Wortman, 1989, 21). Finally, the reforms were intended to *increase* fiscal revenue and expand state capacity. If liberal reform were truly beneficial for peasant income, maintaining the commune seems surprising. However, the government’s belief that liberal reform could have shrunk agricultural output and caused land alienation suggests that uncertainty played a crucial role in the post-1861 mixture.

¹⁹For example, “[a]dditional legislation in 1893 explicitly forbade all sales of allotment land to non-peasants” and required a two-thirds majority of communal assembly votes to redeem or release landholdings and increased the portion of communal assembly votes (to two-thirds) required for a household to obtain release from their share or to redeem their land individually (Nafziger, 2010, 393).